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GENERAL STRUCTURE OF THE GRAVITATIONAL EQUATIONS OF MOTION IN CONFORMAL WEYL GRAVITY

DEMOSTHENES KAZANAS
Laboratory for High Energy Astrophysics, NASA/Goddard Space Flight Center, Greenbelt, MD 20771
AND
PHILIP D. MANNHEIM
Department of Physics, University of Connecticut, Storrs, CT 06269
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ABSTRACT

We present a general method for determining the structure of the gravitational equations of motion in the currently experimentally viable fourth-order theory of gravity based on local conformal Weyl invariance of the gravitational action and give the explicit structure for these equations in the case where the geometry is described by a time-dependent, spherically symmetric line element.

Subject headings: cosmology — gravitation

1. INTRODUCTION

In a recent series of papers (Mannheim & Kazanas 1989a (hereafter Paper I); Mannheim & Kazanas 1989b; Mannheim 1990; Kazanas & Mannheim 1990; Mannheim & Kazanas 1990a, b), we have embarked on a study of a locally conformal invariant theory of gravity and have shown that it could potentially replace the standard Einstein theory of gravity. The theory we presented appears to possess all the successful features of Einstein gravity in the case of an isolated, static, spherically symmetric gravitating source, and at the same time it naturally provides a potential explanation for both the galactic rotation curve–dark matter problem (Paper I) and the cosmological constant problem (Mannheim 1990). In particular, in Paper I we studied Weyl gravity in the case where the line element was static and spherically symmetric, viz.,

$$ds^2 = -B(r)dt^2 + A(r)dr^2 + r^2 d\Omega \tag{1.1}$$

and obtained an exact vacuum solution of the form

$$A^{-1}(r) = B(r) = 1 - \frac{\beta(2-3\beta\gamma)}{r} - 3\beta\gamma + \gamma r - kr^2 . \tag{1.2}$$

As far as we know, this is the first exact solution obtained in theories of gravity based on fourth-order rather than second-order equations of motion. Indeed, because the Weyl theory is two orders higher in derivatives of the metric than the Einstein theory, its static vacuum solution possesses two extra constants of integration, labeled γ and k in equation (1.2), in addition to the familiar Schwarzschild term (parameterized here by β). When nonzero, the integration constant k characterizes a background de Sitter (or anti-de Sitter) geometry, so that in Weyl gravity the de Sitter geometry is a vacuum ($T^{\mu\nu} = 0$) solution and can thus occur even when there is no cosmological term (Mannheim 1990), while the integration constant γ effectively measures all departures of the local geometry associated with the metric of equation (1.2) from that of the Einstein theory. When the contribution of the $(\gamma r - 3\beta\gamma)$ term to the metric is small, the Weyl theory becomes indistinguishable from the Einstein one, and, correspondingly, as this contribution gets larger, the two theories eventually become experimentally distinct. Moreover, we argued in Paper I that this additional term may actually be important on galactic length scales (thus leaving the standard static tests of Einstein gravity on stellar scales intact) where it could then prevent galactic rotational velocities from falling with distance. Thus Weyl gravity could potentially provide a solution to the longstanding galactic rotation curve problem without the need for the introduction of any unobserved dark matter. Conformal Weyl gravity would thus, at present, seem to be a viable theory of gravity, and in order to further confront it with experiment, it becomes necessary to study the general structure of its gravitational equations of motion in greater depth. In this purely mathematical paper we describe the general calculational method we have used in our papers and determine the explicit structure of the equations of motion in a case of much practical interest, namely the gravitational field equations associated with a time-dependent, spherically symmetric geometry. We believe that the equations that we present here will provide a useful starting point for future investigations.

At this point the reader may be wondering how our work could possibly be of interest, since there seems to be a broad consensus in the community that the question of what the correct theory of gravity is has long since been resolved in favor Einstein (at least at the classical level). To respond to this, we note that what has essentially been shown experimentally is that gravity is indeed a purely metric theory with test particles being required to move on geodesics. Now, in order to put a test particle on a geodesic, it is only necessary for the gravitational action to be a general coordinate scalar (a feature shared by both the Weyl and Einstein actions). Then, the appropriate Bianchi identities, viz.

$$(\delta I/\delta g_{\mu\nu})_{;\mu} = 0 \tag{1.3}$$

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ensure that the energy momentum tensor of a test particle is automatically covariantly conserved, thereby enforcing the equivalence principle for test particles and requiring them to move on geodesics which are independent of their own mass. Moreover, the standard static tests of general relativity have (to a very high degree) established that the structure of the geometry which fixes the geodesics in the vicinity of the sun is given by the Schwarzschild line element. As can be seen from the structure of the line element of equation (1.2), experiment would thus also seem to permit fourth-order Weyl gravity on stellar distance scales.[1] Definitive differences between Weyl and Einstein gravity would then only become significant on much larger scales, with the appropriate scales themselves only being establishable by observations.[2] Thus even at this late date, there would still seem to be some freedom left in determining the correct theory of gravity, and in this paper we develop some of the formalism which may eventually enable us to obtain an experimental resolution of the issue.

2. GENERAL STRUCTURE OF WEYL GRAVITY

By itself the requirement that the gravitational action be a general coordinate scalar still leaves open a great deal of freedom in choosing the correct form for the action. In his original formulation of general relativity, Einstein himself made a specific choice by demanding that the gravitational field equations contain only second-order derivatives of the metric. This additional constraint then fixes the action to simply be linear in the Ricci scalar $R^{\alpha}{}_{\alpha}$. (This choice itself is not actually the simplest possible choice one could make, since a nontrivial gravitational theory can be obtained by having the action be proportional to a pure spacetime constant, the infamous cosmological term). In order for us to make a choice here, we shall appeal not to simplicity but rather to an invariance principle. Taking note of the fact that the principle of local gauge invariance severely restricts the structure of possible Lorentz invariant actions in flat space time theories, we follow Weyl and impose the quite analogous principle of local conformal invariance as the requisite invariance principle in the curved space time case. This principle requires the gravitational action to remain invariant under any and all local stretchings

$$g_{\mu\nu}(x) \rightarrow \Omega^2(x) g_{\mu\nu}(x) \tag{2.1}$$

of the geometry. Under these stretchings, the conformal Weyl tensor (Weyl 1918)

$$C_{\lambda\mu\nu\kappa} = R_{\lambda\mu\nu\kappa} - \tfrac{1}{2}(g_{\lambda\nu}R_{\mu\kappa} - g_{\lambda\kappa}R_{\mu\nu} - g_{\mu\nu}R_{\lambda\kappa} + g_{\mu\kappa}R_{\lambda\nu}) + \tfrac{1}{6} R^{\alpha}{}_{\alpha}(g_{\lambda\nu}g_{\mu\kappa} - g_{\lambda\kappa}g_{\mu\nu}) \tag{2.2}$$

transforms as

$$C_{\lambda\mu\nu\kappa} \rightarrow \Omega^2(x) C_{\lambda\mu\nu\kappa} \tag{2.3}$$

even though the Riemann and Ricci tensors, $R_{\lambda\mu\nu\kappa}$ and $R_{\lambda\nu}$, respectively, acquire complicated derivatives of the local stretching function $\Omega(x)$. Consequently the Weyl action I_W (whose coefficient α is purely dimensionless in four spacetime dimensions alone), viz.

$$I_W = -\alpha \int d^4x (-g)^{1/2} C_{\lambda\mu\nu\kappa} C^{\lambda\mu\nu\kappa}, \tag{2.4}$$

is completely locally conformal invariant and represents the unique four-dimensional general coordinate scalar action with the requisite properties. Being unique, the action of equation (2.4) is thus a very attractive object theoretically, but being fourth-order in the derivatives of the metric is not as easy to deal with as the much simpler Einstein action.

Taking note of the fact (Lanczos 1938) that the quantity

$$(-g)^{1/2}[R_{\lambda\mu\nu\kappa}R^{\lambda\mu\nu\kappa} - 4R_{\mu\kappa}R^{\mu\kappa} + (R^{\alpha}{}_{\alpha})^2]$$

is a total divergence enables us to simplify the Weyl action somewhat and rewrite it as

$$I_W = -2\alpha \int d^4x (-g)^{1/2}[R_{\mu\kappa}R^{\mu\kappa} - \frac{1}{3}(R^{\alpha}{}_{\alpha})^2]. \tag{2.5}$$

Variation of this action with respect to $g_{\mu\nu}(x)$ then gives

$$(-g)^{-1/2} g_{\mu\alpha} g_{\nu\beta} \frac{\delta I_W}{\delta g_{\alpha\beta}} = -2\alpha \left[W^{(2)}_{\mu\nu} - \frac{1}{3} W^{(1)}_{\mu\nu} \right] = -2\alpha W_{\mu\nu} \tag{2.6}$$

where (see, e.g., Mannheim 1990)

$$W^{(1)}_{\mu\nu} = 2g_{\mu\nu}(R^{\alpha}{}_{\alpha})^{;\beta}{}_{;\beta} - 2(R^{\alpha}{}_{\alpha})_{;\mu;\nu} - 2R^{\alpha}{}_{\alpha}R_{\mu\nu} + \tfrac{1}{2} g_{\mu\nu}(R^{\alpha}{}_{\alpha})^2$$

$$W^{(2)}_{\mu\nu} = \tfrac{1}{2} g_{\mu\nu}(R^{\alpha}{}_{\alpha})^{;\beta}{}_{;\beta} + R_{\mu\nu}{}^{;\beta}{}_{;\beta} - R_{\mu}{}^{\beta}{}_{;\nu;\beta} - R_{\nu}{}^{\beta}{}_{;\mu;\beta} - 2R_{\mu\beta}R_{\nu}{}^{\beta} + \tfrac{1}{2} g_{\mu\nu} R_{\alpha\beta}R^{\alpha\beta} \tag{2.7}$$

The tensor $W^{\mu\nu}$ is automatically covariantly conserved and, because of the underlying conformal invariance, is also identically covariantly traceless. Additionally, since the action is locally conformal invariant $W^{\mu\nu}$ transforms as

$$W^{\mu\nu} \rightarrow \Omega^{-6}(x) W^{\mu\nu} \tag{2.8}$$

under the local transformations of equation (2.1) on the metric. With the use of this $W^{\mu\nu}$ the gravitational equations of motion become

$$2\alpha W^{\mu\nu} - \tfrac{1}{2} T^{\mu\nu} = 0 \tag{2.9}$$

[1] Beyond the standard static tests of general relativity, there is so far only one other kind of observational test available, namely that associated with the decay of the orbit of the binary pulsar PSR 1913+16. Even though our theory does indeed agree with Einstein gravity on binary pulsar length scales in the static, spherically symmetric limit, because of the complexity of the structure of time-dependent, non–spherically symmetric Weyl gravity, we have not yet been able to address this question in our theory. We are well aware of the significance of this test to the ultimate viability of Weyl gravity.

[2] Independent of our own interest in the galactic rotation curve problem both here and in Paper I, we note in passing that the very need for large amounts of so far unobserved dark matter in the standard gravitational theory may actually be indicative of a failure of the Einstein-Newton theory on galactic length scales.

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in the presence of an energy-momentum tensor $T^{\mu\nu}$, with equations (2.9) then being the candidate gravitational equations which are to replace the Einstein equations in conformal Weyl gravity.

Rather than try to evaluate $W^{\mu\nu}$ as given by equation (2.7) for any given geometrical line element of interest, we found it simpler in the static case considered in Paper I to evaluate $W^{\mu\nu}$ directly via equation (2.6), and we now present the extension of the approach of Paper I to the case of the general time-dependent, spherically symmetric line element, which we take to be given by the initially too general, but nonetheless very convenient, form

$$ds^2 = -B(r, t)dt^2 + A(r, t)dr^2 + C(r, t)d\Omega\,. \quad (2.10)$$

Instead of doing the functional variation required for equation (2.6) formally, we can do it directly from its definition. In terms of the Lagrangian density K, viz.

$$K = (-g)^{1/2}[R_{\mu\kappa}R^{\mu\kappa} - \tfrac{1}{3}(R^{\alpha}{}_{\alpha})^2] \quad (2.11)$$

we find that Euler-Lagrange variation yields for W^{tt} the expression

$$-(-g)^{1/2}W^{tt} = -\frac{1}{2\alpha}\frac{\delta I_W}{\delta B} = \frac{\partial K}{\partial B} - \frac{\partial}{\partial r}\left(\frac{\partial K}{\partial B'}\right) - \frac{\partial}{\partial t}\left(\frac{\partial K}{\partial \dot{B}}\right) + \frac{\partial}{\partial r}\frac{\partial}{\partial t}\left(\frac{\partial K}{\partial \dot{B}'}\right) + \frac{\partial^2}{\partial r^2}\left(\frac{\partial K}{\partial B''}\right) + \frac{\partial^2}{\partial t^2}\left(\frac{\partial K}{\partial \ddot{B}}\right) \quad (2.12)$$

with W^{rr} being given analogously via

$$-2\alpha(-g)^{1/2}W^{rr} = \frac{\delta I_W}{\delta A}\,. \quad (2.13)$$

Since $W^{\mu\nu}$ is traceless, we do not need to calculate $W^{\theta\theta}$ explicitly (nor of course $W^{\phi\phi}$ which is given by $W^{\theta\theta}/\sin^2\theta$ in a spherically symmetric theory) but instead we can obtain $W^{\theta\theta}$ from the trace condition

$$-BW^{tt} + AW^{rr} + 2CW^{\theta\theta} = 0\,. \quad (2.14)$$

In the spherically symmetric case of interest the only remaining nonzero element of $W^{\mu\nu}$ is W^{rt}. To obtain W^{rt} we must momentarily consider the most general spherically symmetric line element possible, viz.

$$ds^2 = -B(r, t)dt^2 + E(r, t)drdt + A(r, t)dr^2 + C(r, t)d\Omega \quad (2.15)$$

and evaluate W^{rt} via

$$-2\alpha(-g)^{1/2}W^{rt} = \frac{\delta I_W}{\delta E}\,. \quad (2.16)$$

Setting $E(r, t)$ equal to zero in equation (2.16) after the variation then yields us the tensor component W^{rt} associated with the line element of equation (2.10).

As we can see from equation (2.12) and its analogs, we may determine the nonzero elements of $W^{\mu\nu}$ by ordinary differentiation, without the need to resort to the covariant differentiations required for equation (2.7). Since the evaluation of equation (2.12) only requires ordinary differentiation, we can do the entire calculation symbolically on a computer. We have performed the actual calculations using the computer software package MACSYMA and list the actual expressions that we have obtained for the relevant elements of $W^{\mu\nu}$ in Appendix A. As a check on our calculations we note that the covariant conservation of the tensor $W^{\mu\nu}$, viz.

$$W^{\mu\nu}{}_{;\mu} = (-g)^{-1/2}\partial_{\mu}[(-g)^{1/2}W^{\mu\nu}] + \Gamma^{\nu}_{\mu\lambda}W^{\mu\lambda} = 0 \quad (2.17)$$

and the tracelessness condition of equation (2.14) yield the following two relations which we have verified that the tensor $W^{\mu\nu}$ does indeed obey:

$$\partial_t W^{tt} + \partial_r W^{rt} + \left(\frac{\dot{A}}{2A} + \frac{\dot{B}}{B} + \frac{3\dot{C}}{2C}\right)W^{tt} + \left(\frac{A'}{2A} + \frac{3B'}{2B} + \frac{C'}{C}\right)W^{rt} + \left(\frac{\dot{A}}{2B} - \frac{A\dot{C}}{2BC}\right)W^{rr} = 0 \quad (2.18)$$

and

$$\partial_t W^{rt} + \partial_r W^{rr} + \left(\frac{3\dot{A}}{2A} + \frac{\dot{B}}{2B} + \frac{\dot{C}}{C}\right)W^{rt} + \left(\frac{A'}{A} + \frac{B'}{2B} + \frac{3C'}{2C}\right)W^{rr} + \left(\frac{B'}{2A} - \frac{BC'}{2AC}\right)W^{tt} = 0\,. \quad (2.19)$$

An even more sensitive test of the correctness of our results is to note that under the combined transformation

$$\begin{aligned} A(r, t) &\to \Omega^2(r, t)A(r, t) \\ B(r, t) &\to \Omega^2(r, t)B(r, t) \\ C(r, t) &\to \Omega^2(r, t)C(r, t)\,, \end{aligned} \quad (2.20)$$

where $\Omega(r, t)$ is an arbitrary function of r and t, $W^{\mu\nu}$ must transform (and is in fact found to transform) according to equation (2.8) with all derivatives of $\Omega(r, t)$ being canceled identically. (It is because we carried all of A, B, and C in our general $g_{\mu\nu}$ that we are able to make this conclusive test.) We are thus confident that our general expressions for the nonzero elements of $W^{\mu\nu}$ are indeed correct. Now that we have our too general $W^{\mu\nu}$ we can make specific choices for values for the components of $g_{\mu\nu}$ and evaluate $W^{\mu\nu}$ in various standard cases of physical interest. The expressions for each of these varying cases of interest are all tabulated in Appendix B. As noted in Paper I, despite the apparent complexity of our expressions, when the conformal invariance associated with equation (2.20) was fully exploited in the static case, the entire structure of $W^{\mu\nu}$ was found to reduce to quite a simple form which then admitted of the line element of equation (1.2) as its exact vacuum solution. Thus gravity theories based on $W^{\mu\nu}$ may prove

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to be more tractable than the formulae given in the appendices at first suggest, and only time will tell whether they may even be preferred by observation.

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APPENDIX A

In this appendix we present expressions for the components of the gravitational tensor $W^{\mu\nu}$ and the conformal Weyl tensor $C_{\lambda\mu\nu\kappa}$ which are associated with the time-dependent spherically symmetric line element given in equation (2.10). In terms of the quantity Q defined by

$$Q = 2AB^2CC_{rr} - 2AB^2C_r^2 + ABB_rCC_r - A_rB^2CC_r - 2A^2BCC_{tt} + 2A^2BC_t^2 + A^2B_tCC_t - AA_tBCC_t$$
$$- 2ABB_{rr}C^2 + AB_r^2C^2 + A_rBB_rC^2 - AA_tB_tC^2 + 2AA_{tt}BC^2 - A_t^2BC^2 + 4A^2B^2C ,$$

the nonvanishing components of the Weyl tensor are given by

$$C_{r\theta r\theta} = -\frac{Q}{24AB^2C}, \qquad C_{r\phi r\phi} = \sin^2\theta C_{r\theta r\theta},$$

$$C_{rtrt} = -\frac{Q}{12ABC^2}, \qquad C_{\theta\phi\theta\phi} = \frac{Q\sin^2\theta}{12A^2B^2},$$

$$C_{\theta t\theta t} = \frac{Q}{24A^2BC}, \qquad C_{\phi t\phi t} = \sin^2\theta C_{\theta t\theta t}.$$

We denote the relevant components of $W^{\mu\nu}$ by W^{rr}, W^{rt}, and W^{tt}, and they are found to take the forms which follow below.

```
W^rr=

(8 A^2 B^5 C^2 C_r C_r r r - 8 A^2 B^4 B_r C^3 C_r r r - 4 A^2 B^5 C^2 (C_r r)^2

- 8 A^2 B^5 C (C_r)^2 C_r r + 16 A^2 B^4 B_r C^2 C_r C_r r - 8 A A_r B^5 C^2 C_r C_r r

+ 8 A^3 B^4 C^2 C_t t C_r r - 8 A^3 B^4 C (C_t)^2 C_r r - 4 A^3 B^3 B_t C^2 C_t C_r r

- 4 A^2 A_t B^4 C^2 C_t C_r r + 8 A^2 B^4 B_r r C^3 C_r r - 8 A^2 B^3 (B_r)^2 C^3 C_r r

+ 8 A A_r B^4 B_r C^3 C_r r + 12 A^2 A_t B^3 B_t C^3 C_r r - 24 A^2 A_t t B^4 C^3 C_r r

+ 36 A (A_t)^2 B^4 C^3 C_r r + 4 A^2 B^5 (C_r)^4 - 4 A^2 B^4 B_r C (C_r)^3

+ 4 A A_r B^5 C (C_r)^3 + 8 A^3 B^4 C C_t t (C_r)^2 - 24 A^3 B^4 (C_t)^2 (C_r)^2

- 4 A^3 B^3 B_t C C_t (C_r)^2 - 28 A^2 A_t B^4 C C_t (C_r)^2 - 12 A^2 B^4 B_r r C^2 (C_r)^2

+ 3 A^2 B^3 (B_r)^2 C^2 (C_r)^2 - 2 A A_r B^4 B_r C^2 (C_r)^2 - 16 A^2 A_t B^3 B_t C^2 (C_r)^2

- 4 A A_r r B^5 C^2 (C_r)^2 + 7 (A_r)^2 B^5 C^2 (C_r)^2 + 32 A^2 A_t t B^4 C^2 (C_r)^2
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$$
\begin{aligned}
&- 40\,A\,(A_t)^2\,B^4\,C^2\,(C_r)^2 - 40\,A^3\,B^4\,C^2\,C_{t\,t\,r}\,C_r + 12\,A^3\,B^3\,B_r\,C^2\,C_{t\,t}\,C_r \\
&- 4\,A^2\,A_r\,B^4\,C^2\,C_{t\,t}\,C_r + 64\,A^3\,B^4\,C\,C_t\,C_{t\,r}\,C_r + 20\,A^3\,B^3\,B_t\,C^2\,C_{t\,r}\,C_r \\
&+ 60\,A^2\,A_t\,B^4\,C^2\,C_{t\,r}\,C_r - 4\,A^3\,B^3\,B_r\,C\,(C_t)^2\,C_r + 4\,A^2\,A_r\,B^4\,C\,(C_t)^2\,C_r \\
&- 14\,A^3\,B^2\,B_t\,B_r\,C^2\,C_t\,C_r + 2\,A^2\,A_t\,B^3\,B_r\,C^2\,C_t\,C_r + 8\,A^3\,B^3\,B_{t\,r}\,C^2\,C_t\,C_r \\
&+ 2\,A^2\,A_r\,B^3\,B_t\,C^2\,C_t\,C_r + 10\,A\,A_t\,A_r\,B^4\,C^2\,C_t\,C_r - 8\,A^2\,A_{t\,r}\,B^4\,C^2\,C_t\,C_r \\
&- 8\,A^2\,B^4\,B_{r\,r\,r}\,C^3\,C_r + 24\,A^2\,B^3\,B_r\,B_{r\,r}\,C^3\,C_r + 8\,A\,A_r\,B^4\,B_{r\,r}\,C^3\,C_r \\
&- 10\,A^2\,B^2\,(B_r)^3\,C^3\,C_r - 8\,A\,A_r\,B^3\,(B_r)^2\,C^3\,C_r - 8\,A^3\,B^2\,B_{t\,t}\,B_r\,C^3\,C_r \\
&+ 20\,A^3\,B\,(B_t)^2\,B_r\,C^3\,C_r + 34\,A^2\,A_t\,B^2\,B_t\,B_r\,C^3\,C_r + 8\,A\,A_{r\,r}\,B^4\,B_r\,C^3\,C_r \\
&- 14\,(A_r)^2\,B^4\,B_r\,C^3\,C_r - 28\,A^2\,A_{t\,t}\,B^3\,B_r\,C^3\,C_r + 26\,A\,(A_t)^2\,B^3\,B_r\,C^3\,C_r \\
&+ 8\,A^3\,B^3\,B_{t\,t\,r}\,C^3\,C_r - 20\,A^3\,B^2\,B_t\,B_{t\,r}\,C^3\,C_r - 20\,A^2\,A_t\,B^3\,B_{t\,r}\,C^3\,C_r \\
&- 6\,A\,A_t\,A_r\,B^3\,B_t\,C^3\,C_r + 12\,A\,A_{t\,t}\,A_r\,B^4\,C^3\,C_r - 42\,(A_t)^2\,A_r\,B^4\,C^3\,C_r \\
&+ 24\,A\,A_t\,A_{t\,r}\,B^4\,C^3\,C_r - 16\,A^4\,B^3\,C^3\,C_{t\,t\,t\,t} + 24\,A^4\,B^3\,C^2\,C_t\,C_{t\,t\,t} \\
&+ 48\,A^4\,B^2\,B_t\,C^3\,C_{t\,t\,t} - 8\,A^3\,A_t\,B^3\,C^3\,C_{t\,t\,t} + 16\,A^3\,B^4\,C^3\,C_{t\,t\,r\,r} \\
&+ 16\,A^3\,B^3\,B_r\,C^3\,C_{t\,t\,r} - 8\,A^2\,A_r\,B^4\,C^3\,C_{t\,t\,r} + 28\,A^4\,B^3\,C^2\,(C_{t\,t})^2 \\
&- 56\,A^4\,B^3\,C\,(C_t)^2\,C_{t\,t} - 64\,A^4\,B^2\,B_t\,C^2\,C_t\,C_{t\,t} - 8\,A^3\,A_t\,B^3\,C^2\,C_t\,C_{t\,t} \\
&- 24\,A^3\,B^3\,B_{r\,r}\,C^3\,C_{t\,t} + 4\,A^3\,B^2\,(B_r)^2\,C^3\,C_{t\,t} + 12\,A^2\,A_r\,B^3\,B_r\,C^3\,C_{t\,t} \\
&+ 32\,A^4\,B^2\,B_{t\,t}\,C^3\,C_{t\,t} - 76\,A^4\,B\,(B_t)^2\,C^3\,C_{t\,t} + 8\,A^3\,A_t\,B^2\,B_t\,C^3\,C_{t\,t} \\
&+ 8\,A^3\,A_{t\,t}\,B^3\,C^3\,C_{t\,t} + 4\,A^2\,(A_t)^2\,B^3\,C^3\,C_{t\,t} + 8\,A^3\,B^4\,C^2\,C_t\,C_{t\,r\,r} \\
&- 8\,A^3\,B^3\,B_t\,C^3\,C_{t\,r\,r} - 32\,A^2\,A_t\,B^4\,C^3\,C_{t\,r\,r} - 32\,A^3\,B^4\,C^2\,(C_{t\,r})^2 \\
&- 12\,A^3\,B^3\,B_r\,C^2\,C_t\,C_{t\,r} - 4\,A^2\,A_r\,B^4\,C^2\,C_t\,C_{t\,r} - 24\,A^3\,B^2\,B_t\,B_r\,C^3\,C_{t\,r} \\
&- 12\,A^2\,A_t\,B^3\,B_r\,C^3\,C_{t\,r} + 16\,A^3\,B^3\,B_{t\,r}\,C^3\,C_{t\,r} + 4\,A^2\,A_r\,B^3\,B_t\,C^3\,C_{t\,r} \\
&+ 32\,A\,A_t\,A_r\,B^4\,C^3\,C_{t\,r} - 16\,A^2\,A_{t\,r}\,B^4\,C^3\,C_{t\,r} + 20\,A^4\,B^3\,(C_t)^4
\end{aligned}
$$

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$$+ 28\,A^4\,B^2\,B_t\,C\,(C_t)^3 + 4\,A^3\,A_t\,B^3\,C\,(C_t)^3 + 8\,A^3\,B^2\,(B_r)^2\,C^2\,(C_t)^2$$

$$- 12\,A^4\,B^2\,B_{t\,t}\,C^2\,(C_t)^2 + 31\,A^4\,B\,(B_t)^2\,C^2\,(C_t)^2 + 6\,A^3\,A_t\,B^2\,B_t\,C^2\,(C_t)^2$$

$$- 4\,A^3\,A_{t\,t}\,B^3\,C^2\,(C_t)^2 + 3\,A^2\,(A_t)^2\,B^3\,C^2\,(C_t)^2 + 52\,A^3\,B^2\,B_t\,B_{r\,r}\,C^3\,C_t$$

$$+ 36\,A^2\,A_t\,B^3\,B_{r\,r}\,C^3\,C_t - 38\,A^3\,B\,B_t\,(B_r)^2\,C^3\,C_t - 22\,A^2\,A_t\,B^2\,(B_r)^2\,C^3\,C_t$$

$$+ 36\,A^3\,B^2\,B_{t\,r}\,B_r\,C^3\,C_t - 26\,A^2\,A_r\,B^2\,B_t\,B_r\,C^3\,C_t - 42\,A\,A_t\,A_r\,B^3\,B_r\,C^3\,C_t$$

$$+ 24\,A^2\,A_{t\,r}\,B^3\,B_r\,C^3\,C_t + 8\,A^4\,B^2\,B_{t\,t\,t}\,C^3\,C_t - 52\,A^4\,B\,B_t\,B_{t\,t}\,C^3\,C_t$$

$$- 12\,A^3\,A_t\,B^2\,B_{t\,t}\,C^3\,C_t - 40\,A^3\,B^3\,B_{t\,r\,r}\,C^3\,C_t + 20\,A^2\,A_r\,B^3\,B_{t\,r}\,C^3\,C_t$$

$$+ 56\,A^4\,(B_t)^3\,C^3\,C_t + 26\,A^3\,A_t\,B\,(B_t)^2\,C^3\,C_t - 52\,A^3\,A_{t\,t}\,B^2\,B_t\,C^3\,C_t$$

$$+ 24\,A^2\,(A_t)^2\,B^2\,B_t\,C^3\,C_t + 32\,A^3\,A_{t\,t\,t}\,B^3\,C^3\,C_t - 52\,A^2\,A_t\,A_{t\,t}\,B^3\,C^3\,C_t$$

$$+ 22\,A\,(A_t)^3\,B^3\,C^3\,C_t + 8\,A^2\,B^3\,B_r\,B_{r\,r\,r}\,C^4 - 4\,A^2\,B^3\,(B_{r\,r})^2\,C^4$$

$$- 12\,A^2\,B^2\,(B_r)^2\,B_{r\,r}\,C^4 - 8\,A\,A_r\,B^3\,B_r\,B_{r\,r}\,C^4 + 16\,A^3\,B^2\,B_{t\,t}\,B_{r\,r}\,C^4$$

$$- 40\,A^3\,B\,(B_t)^2\,B_{r\,r}\,C^4 - 44\,A^2\,A_t\,B^2\,B_t\,B_{r\,r}\,C^4 + 24\,A^2\,A_{t\,t}\,B^3\,B_{r\,r}\,C^4$$

$$- 36\,A\,(A_t)^2\,B^3\,B_{r\,r}\,C^4 + 7\,A^2\,B\,(B_r)^4\,C^4 + 6\,A\,A_r\,B^2\,(B_r)^3\,C^4$$

$$- 16\,A^3\,B\,B_{t\,t}\,(B_r)^2\,C^4 + 56\,A^3\,(B_t)^2\,(B_r)^2\,C^4 + 30\,A^2\,A_t\,B\,B_t\,(B_r)^2\,C^4$$

$$- 4\,A\,A_{r\,r}\,B^3\,(B_r)^2\,C^4 + 7\,(A_r)^2\,B^3\,(B_r)^2\,C^4 - 4\,A^2\,A_{t\,t}\,B^2\,(B_r)^2\,C^4$$

$$+ 14\,A\,(A_t)^2\,B^2\,(B_r)^2\,C^4 + 16\,A^3\,B^2\,B_{t\,t\,r}\,B_r\,C^4 - 8\,A^2\,A_r\,B^2\,B_{t\,t}\,B_r\,C^4$$

$$- 72\,A^3\,B\,B_t\,B_{t\,r}\,B_r\,C^4 - 28\,A^2\,A_t\,B^2\,B_{t\,r}\,B_r\,C^4 + 20\,A^2\,A_r\,B\,(B_t)^2\,B_r\,C^4$$

$$+ 38\,A\,A_t\,A_r\,B^2\,B_t\,B_r\,C^4 - 16\,A^2\,A_{t\,r}\,B^2\,B_t\,B_r\,C^4 - 12\,A\,A_{t\,t}\,A_r\,B^3\,B_r\,C^4$$

$$+ 42\,(A_t)^2\,A_r\,B^3\,B_r\,C^4 - 24\,A\,A_t\,A_{t\,r}\,B^3\,B_r\,C^4 - 8\,A^3\,A_t\,B^2\,B_{t\,t\,t}\,C^4$$

$$- 16\,A^3\,B^3\,B_{t\,t\,r\,r}\,C^4 + 8\,A^2\,A_r\,B^3\,B_{t\,t\,r}\,C^4 + 52\,A^3\,A_t\,B\,B_t\,B_{t\,t}\,C^4$$

$$- 32\,A^3\,A_{t\,t}\,B^2\,B_{t\,t}\,C^4 + 24\,A^2\,(A_t)^2\,B^2\,B_{t\,t}\,C^4 + 40\,A^3\,B^2\,B_t\,B_{t\,r\,r}\,C^4$$

$$+ 32\,A^2\,A_t\,B^3\,B_{t\,r\,r}\,C^4 + 16\,A^3\,B^2\,(B_{t\,r})^2\,C^4 - 20\,A^2\,A_r\,B^2\,B_t\,B_{t\,r}\,C^4$$

$$- 32\,A\,A_t\,A_r\,B^3\,B_{t\,r}\,C^4 + 16\,A^2\,A_{t\,r}\,B^3\,B_{t\,r}\,C^4 - 56\,A^3\,A_t\,(B_t)^3\,C^4$$

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$$+ 76 A^3 A_{t\,t} B (B_t)^2 C^4 - 57 A^2 (A_t)^2 B (B_t)^2 C^4 - 48 A^3 A_{t\,t\,t} B^2 B_t C^4$$

$$+ 108 A^2 A_t A_{t\,t} B^2 B_t C^4 - 58 A (A_t)^3 B^2 B_t C^4 + 16 A^3 A_{t\,t\,t\,t} B^3 C^4$$

$$- 48 A^2 A_t A_{t\,t\,t} B^3 C^4 - 36 A^2 (A_{t\,t})^2 B^3 C^4 + 116 A (A_t)^2 A_{t\,t} B^3 C^4$$

$$- 49 (A_t)^4 B^3 C^4 + 16 A^4 B^5 C^2)/(48 A^5 B^5 C^4)$$

$$W^{rt} =$$

$$- (2 A^3 B^4 C^2 C_t C_{r\,r\,r} - 6 A^2 A_t B^4 C^3 C_{r\,r\,r} + 4 A^3 B^4 C C_t C_r C_{r\,r}$$

$$+ 10 A^2 A_t B^4 C^2 C_r C_{r\,r} - 8 A^3 B^4 C^2 C_{t\,r} C_{r\,r} + A^3 B^3 B_r C^2 C_t C_{r\,r}$$

$$- 3 A^2 A_r B^4 C^2 C_t C_{r\,r} - 2 A^3 B^2 B_t B_r C^3 C_{r\,r} - A^2 A_t B^3 B_r C^3 C_{r\,r}$$

$$+ 2 A^3 B^3 B_{t\,r} C^3 C_{r\,r} + 15 A A_t A_r B^4 C^3 C_{r\,r} - 6 A^2 A_{t\,r} B^4 C^3 C_{r\,r}$$

$$- 4 A^3 B^4 C_t (C_r)^3 - 4 A^2 A_t B^4 C (C_r)^3 + 8 A^3 B^4 C C_{t\,r} (C_r)^2$$

$$- 2 A^3 B^3 B_r C C_t (C_r)^2 - 2 A^2 A_r B^4 C C_t (C_r)^2 - A^3 B^2 B_t B_r C^2 (C_r)^2$$

$$- 3 A^2 A_t B^3 B_r C^2 (C_r)^2 + A^3 B^3 B_{t\,r} C^2 (C_r)^2 - 8 A A_t A_r B^4 C^2 (C_r)^2$$

$$+ 3 A^2 A_{t\,r} B^4 C^2 (C_r)^2 - 2 A^4 B^3 C^2 C_{t\,t\,t} C_r - 4 A^4 B^3 C C_t C_{t\,t} C_r$$

$$+ 3 A^4 B^2 B_t C^2 C_{t\,t} C_r - A^3 A_t B^3 C^2 C_{t\,t} C_r - 6 A^3 B^4 C^2 C_{t\,r\,r} C_r$$

$$+ 5 A^3 B^3 B_r C^2 C_{t\,r} C_r + 7 A^2 A_r B^4 C^2 C_{t\,r} C_r + 4 A^4 B^3 (C_t)^3 C_r$$

$$+ 2 A^4 B^2 B_t C (C_t)^2 C_r + 2 A^3 A_t B^3 C (C_t)^2 C_r - 3 A^3 B^3 B_{r\,r} C^2 C_t C_r$$

$$+ A^3 B^2 (B_r)^2 C^2 C_t C_r + A^2 A_r B^3 B_r C^2 C_t C_r + A^4 B^2 B_{t\,t} C^2 C_t C_r$$

$$- 2 A^4 B (B_t)^2 C^2 C_t C_r - A^3 A_t B^2 B_t C^2 C_t C_r - A^2 A_{r\,r} B^4 C^2 C_t C_r$$

$$+ 2 A (A_r)^2 B^4 C^2 C_t C_r + 3 A^3 A_{t\,t} B^3 C^2 C_t C_r - A^2 (A_t)^2 B^3 C^2 C_t C_r$$

$$+ 4 A^3 B^2 B_t B_{r\,r} C^3 C_r + 5 A^2 A_t B^3 B_{r\,r} C^3 C_r - A^3 B B_t (B_r)^2 C^3 C_r$$

$$+ A^3 B^2 B_{t\,r} B_r C^3 C_r - A^2 A_r B^2 B_t B_r C^3 C_r - 4 A A_t A_r B^3 B_r C^3 C_r$$

$$+ 2 A^2 A_{t\,r} B^3 B_r C^3 C_r - 3 A^3 A_t B^2 B_{t\,t} C^3 C_r - 4 A^3 B^3 B_{t\,r\,r} C^3 C_r$$

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$$+ A^2 A_r B^3 B_{t\,r} C^3 C_r + 6 A^3 A_t B (B_t)^2 C^3 C_r - 9 A^3 A_{t\,t} B^2 B_t C^3 C_r$$

$$+ 7 A^2 (A_t)^2 B^2 B_t C^3 C_r + 5 A A_t A_{r\,r} B^4 C^3 C_r - 14 A_t (A_r)^2 B^4 C^3 C_r$$

$$+ 8 A A_{t\,r} A_r B^4 C^3 C_r - 2 A^2 A_{t\,r\,r} B^4 C^3 C_r + 6 A^3 A_{t\,t\,t} B^3 C^3 C_r$$

$$- 14 A^2 A_t A_{t\,t} B^3 C^3 C_r + 7 A (A_t)^3 B^3 C^3 C_r - 4 A^4 B^3 C^3 C_{t\,t\,t\,r}$$

$$+ 6 A^4 B^2 B_r C^3 C_{t\,t\,t} + 6 A^4 B^3 C^2 C_t C_{t\,t\,r} + 6 A^4 B^2 B_t C^3 C_{t\,t\,r}$$

$$+ 8 A^4 B^3 C^2 C_{t\,r} C_{t\,t} - 10 A^4 B^2 B_r C^2 C_t C_{t\,t} - 15 A^4 B B_t B_r C^3 C_{t\,t}$$

$$+ A^3 A_t B^2 B_r C^3 C_{t\,t} + 6 A^4 B^2 B_{t\,r} C^3 C_{t\,t} + 2 A^2 A_t A_r B^3 C^3 C_{t\,t}$$

$$- 2 A^3 A_{t\,r} B^3 C^3 C_{t\,t} + 4 A^3 B^4 C^3 C_{t\,r\,r\,r} - 6 A^2 A_r B^4 C^3 C_{t\,r\,r}$$

$$- 8 A^4 B^3 C (C_t)^2 C_{t\,r} - 7 A^4 B^2 B_t C^2 C_t C_{t\,r} - 5 A^3 A_t B^3 C^2 C_t C_{t\,r}$$

$$- 2 A^3 B^3 B_{r\,r} C^3 C_{t\,r} - A^3 B^2 (B_r)^2 C^3 C_{t\,r} + A^2 A_r B^3 B_r C^3 C_{t\,r}$$

$$+ 2 A^4 B^2 B_{t\,t} C^3 C_{t\,r} - 4 A^4 B (B_t)^2 C^3 C_{t\,r} - A^3 A_t B^2 B_t C^3 C_{t\,r}$$

$$- 2 A^2 A_{r\,r} B^4 C^3 C_{t\,r} + 4 A (A_r)^2 B^4 C^3 C_{t\,r} + 2 A^3 A_{t\,t} B^3 C^3 C_{t\,r}$$

$$+ A^2 (A_t)^2 B^3 C^3 C_{t\,r} + 4 A^4 B^2 B_r C (C_t)^3 + 8 A^4 B B_t B_r C^2 (C_t)^2$$

$$+ 3 A^3 A_t B^2 B_r C^2 (C_t)^2 - 3 A^4 B^2 B_{t\,r} C^2 (C_t)^2 + A^2 A_t A_r B^3 C^2 (C_t)^2$$

$$- A^3 A_{t\,r} B^3 C^2 (C_t)^2 - 6 A^3 B^3 B_{r\,r\,r} C^3 C_t + 14 A^3 B^2 B_r B_{r\,r} C^3 C_t$$

$$+ 9 A^2 A_r B^3 B_{r\,r} C^3 C_t - 7 A^3 B (B_r)^3 C^3 C_t - 7 A^2 A_r B^2 (B_r)^2 C^3 C_t$$

$$- 5 A^4 B B_{t\,t} B_r C^3 C_t + 14 A^4 (B_t)^2 B_r C^3 C_t + 4 A^3 A_t B B_t B_r C^3 C_t$$

$$+ 3 A^2 A_{r\,r} B^3 B_r C^3 C_t - 6 A (A_r)^2 B^3 B_r C^3 C_t - 5 A^3 A_{t\,t} B^2 B_r C^3 C_t$$

$$+ 2 A^4 B^2 B_{t\,t\,r} C^3 C_t - 8 A^4 B B_t B_{t\,r} C^3 C_t - 2 A^3 A_t B^2 B_{t\,r} C^3 C_t$$

$$+ A^2 A_t A_r B^2 B_t C^3 C_t - A^3 A_{t\,r} B^2 B_t C^3 C_t - 4 A^2 A_{t\,t} A_r B^3 C^3 C_t$$

$$+ A (A_t)^2 A_r B^3 C^3 C_t + 4 A^3 A_{t\,t\,r} B^3 C^3 C_t - A^2 A_t A_{t\,r} B^3 C^3 C_t$$

$$+ 4 A^3 B^2 B_t B_{r\,r\,r} C^4 + 6 A^2 A_t B^3 B_{r\,r\,r} C^4 - 18 A^3 B B_t B_r B_{r\,r} C^4$$

$$- 14 A^2 A_t B^2 B_r B_{r\,r} C^4 + 8 A^3 B^2 B_{t\,r} B_{r\,r} C^4 - 6 A^2 A_r B^2 B_t B_{r\,r} C^4$$

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$$- 15\,A\,A_t\,A_r\,B^3\,B_{r\,r}\,C^4 + 6\,A^2\,A_{t\,r}\,B^3\,B_{r\,r}\,C^4 + 14\,A^3\,B_t\,(B_r)^3\,C^4$$

$$+ 7\,A^2\,A_t\,B\,(B_r)^3\,C^4 - 14\,A^3\,B\,B_{t\,r}\,(B_r)^2\,C^4 + 9\,A^2\,A_r\,B\,B_t\,(B_r)^2\,C^4$$

$$+ 12\,A\,A_t\,A_r\,B^2\,(B_r)^2\,C^4 - 5\,A^2\,A_{t\,r}\,B^2\,(B_r)^2\,C^4 + 5\,A^3\,A_t\,B\,B_{t\,t}\,B_r\,C^4$$

$$+ 10\,A^3\,B^2\,B_{t\,r\,r}\,B_r\,C^4 - 9\,A^2\,A_r\,B^2\,B_{t\,r}\,B_r\,C^4 - 14\,A^3\,A_t\,(B_t)^2\,B_r\,C^4$$

$$- 2\,A^2\,A_{r\,r}\,B^2\,B_t\,B_r\,C^4 + 4\,A\,(A_r)^2\,B^2\,B_t\,B_r\,C^4 + 15\,A^3\,A_{t\,t}\,B\,B_t\,B_r\,C^4$$

$$- 12\,A^2\,(A_t)^2\,B\,B_t\,B_r\,C^4 - 5\,A\,A_t\,A_{r\,r}\,B^3\,B_r\,C^4 + 14\,A_t\,(A_r)^2\,B^3\,B_r\,C^4$$

$$- 8\,A\,A_{t\,r}\,A_r\,B^3\,B_r\,C^4 + 2\,A^2\,A_{t\,r\,r}\,B^3\,B_r\,C^4 - 6\,A^3\,A_{t\,t\,t}\,B^2\,B_r\,C^4$$

$$+ 14\,A^2\,A_t\,A_{t\,t}\,B^2\,B_r\,C^4 - 7\,A\,(A_t)^3\,B^2\,B_r\,C^4 - 2\,A^3\,A_t\,B^2\,B_{t\,t\,r}\,C^4$$

$$+ 2\,A^2\,A_t\,A_r\,B^2\,B_{t\,t}\,C^4 - 2\,A^3\,A_{t\,r}\,B^2\,B_{t\,t}\,C^4 - 4\,A^3\,B^3\,B_{t\,r\,r\,r}\,C^4$$

$$+ 6\,A^2\,A_r\,B^3\,B_{t\,r\,r}\,C^4 + 8\,A^3\,A_t\,B\,B_t\,B_{t\,r}\,C^4 + 2\,A^2\,A_{r\,r}\,B^3\,B_{t\,r}\,C^4$$

$$- 4\,A\,(A_r)^2\,B^3\,B_{t\,r}\,C^4 - 6\,A^3\,A_{t\,t}\,B^2\,B_{t\,r}\,C^4 + 5\,A^2\,(A_t)^2\,B^2\,B_{t\,r}\,C^4$$

$$- 4\,A^2\,A_t\,A_r\,B\,(B_t)^2\,C^4 + 4\,A^3\,A_{t\,r}\,B\,(B_t)^2\,C^4 + 6\,A^2\,A_{t\,t}\,A_r\,B^2\,B_t\,C^4$$

$$- 9\,A\,(A_t)^2\,A_r\,B^2\,B_t\,C^4 - 6\,A^3\,A_{t\,t\,r}\,B^2\,B_t\,C^4 + 9\,A^2\,A_t\,A_{t\,r}\,B^2\,B_t\,C^4$$

$$- 4\,A^2\,A_{t\,t\,t}\,A_r\,B^3\,C^4 + 18\,A\,A_t\,A_{t\,t}\,A_r\,B^3\,C^4 - 14\,(A_t)^3\,A_r\,B^3\,C^4$$

$$+ 4\,A^3\,A_{t\,t\,t\,r}\,B^3\,C^4 - 10\,A^2\,A_t\,A_{t\,t\,r}\,B^3\,C^4 - 8\,A^2\,A_{t\,r}\,A_{t\,t}\,B^3\,C^4$$

$$+ 14\,A\,(A_t)^2\,A_{t\,r}\,B^3\,C^4)/(12\,A^5\,B^5\,C^4)$$

$$w^{tt} =$$

$$(16\,A^3\,B^4\,C^3\,C_{r\,r\,r\,r} - 24\,A^3\,B^4\,C^2\,C_r\,C_{r\,r\,r} + 8\,A^3\,B^3\,B_r\,C^3\,C_{r\,r\,r}$$

$$- 48\,A^2\,A_r\,B^4\,C^3\,C_{r\,r\,r} - 28\,A^3\,B^4\,C^2\,(C_{r\,r})^2 + 56\,A^3\,B^4\,C\,(C_r)^2\,C_{r\,r}$$

$$+ 8\,A^3\,B^3\,B_r\,C^2\,C_r\,C_{r\,r} + 64\,A^2\,A_r\,B^4\,C^2\,C_r\,C_{r\,r} - 8\,A^4\,B^3\,C^2\,C_{t\,t}\,C_{r\,r}$$

$$- 8\,A^4\,B^3\,C\,(C_t)^2\,C_{r\,r} + 4\,A^4\,B^2\,B_t\,C^2\,C_t\,C_{r\,r} - 12\,A^3\,A_t\,B^3\,C^2\,C_t\,C_{r\,r}$$

$$- 8\,A^3\,B^3\,B_{r\,r}\,C^3\,C_{r\,r} - 4\,A^3\,B^2\,(B_r)^2\,C^3\,C_{r\,r} - 8\,A^2\,A_r\,B^3\,B_r\,C^3\,C_{r\,r}$$

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$$
\begin{aligned}
&- 12\,A^3\,A_t\,B^2\,B_t\,C^3\,C_{r\,r} - 32\,A^2\,A_{r\,r}\,B^4\,C^3\,C_{r\,r} + 76\,A\,(A_r)^2\,B^4\,C^3\,C_{r\,r} \\
&+ 24\,A^3\,A_{t\,t}\,B^3\,C^3\,C_{r\,r} - 4\,A^2\,(A_t)^2\,B^3\,C^3\,C_{r\,r} - 20\,A^3\,B^4\,(C_r)^4 \\
&- 4\,A^3\,B^3\,B_r\,C\,(C_r)^3 - 28\,A^2\,A_r\,B^4\,C\,(C_r)^3 + 8\,A^4\,B^3\,C\,C_{t\,t}\,(C_r)^2 \\
&+ 24\,A^4\,B^3\,(C_t)^2\,(C_r)^2 - 4\,A^4\,B^2\,B_t\,C\,C_t\,(C_r)^2 + 4\,A^3\,A_t\,B^3\,C\,C_t\,(C_r)^2 \\
&+ 4\,A^3\,B^3\,B_{r\,r}\,C^2\,(C_r)^2 - 3\,A^3\,B^2\,(B_r)^2\,C^2\,(C_r)^2 - 6\,A^2\,A_r\,B^3\,B_r\,C^2\,(C_r)^2 \\
&+ 12\,A^2\,A_{r\,r}\,B^4\,C^2\,(C_r)^2 - 31\,A\,(A_r)^2\,B^4\,C^2\,(C_r)^2 - 8\,A^2\,(A_t)^2\,B^3\,C^2\,(C_r)^2 \\
&- 8\,A^4\,B^3\,C^2\,C_{t\,t\,r}\,C_r + 4\,A^4\,B^2\,B_r\,C^2\,C_{t\,t}\,C_r + 4\,A^3\,A_r\,B^3\,C^2\,C_{t\,t}\,C_r \\
&- 64\,A^4\,B^3\,C\,C_t\,C_{t\,r}\,C_r + 4\,A^4\,B^2\,B_t\,C^2\,C_{t\,r}\,C_r + 12\,A^3\,A_t\,B^3\,C^2\,C_{t\,r}\,C_r \\
&+ 28\,A^4\,B^2\,B_r\,C\,(C_t)^2\,C_r + 4\,A^3\,A_r\,B^3\,C\,(C_t)^2\,C_r - 10\,A^4\,B\,B_t\,B_r\,C^2\,C_t\,C_r \\
&- 2\,A^3\,A_t\,B^2\,B_r\,C^2\,C_t\,C_r + 8\,A^4\,B^2\,B_{t\,r}\,C^2\,C_t\,C_r - 2\,A^3\,A_r\,B^2\,B_t\,C^2\,C_t\,C_r \\
&+ 14\,A^2\,A_t\,A_r\,B^3\,C^2\,C_t\,C_r - 8\,A^3\,A_{t\,r}\,B^3\,C^2\,C_t\,C_r - 32\,A^3\,B^3\,B_{r\,r\,r}\,C^3\,C_r \\
&+ 52\,A^3\,B^2\,B_r\,B_{r\,r}\,C^3\,C_r + 52\,A^2\,A_r\,B^3\,B_{r\,r}\,C^3\,C_r - 22\,A^3\,B\,(B_r)^3\,C^3\,C_r \\
&- 24\,A^2\,A_r\,B^2\,(B_r)^2\,C^3\,C_r + 42\,A^3\,A_t\,B\,B_t\,B_r\,C^3\,C_r + 12\,A^2\,A_{r\,r}\,B^3\,B_r\,C^3\,C_r \\
&- 26\,A\,(A_r)^2\,B^3\,B_r\,C^3\,C_r - 36\,A^3\,A_{t\,t}\,B^2\,B_r\,C^3\,C_r + 22\,A^2\,(A_t)^2\,B^2\,B_r\,C^3\,C_r \\
&- 24\,A^3\,A_t\,B^2\,B_{t\,r}\,C^3\,C_r + 26\,A^2\,A_t\,A_r\,B^2\,B_t\,C^3\,C_r - 20\,A^3\,A_{t\,r}\,B^2\,B_t\,C^3\,C_r \\
&- 8\,A^2\,A_{r\,r\,r}\,B^4\,C^3\,C_r + 52\,A\,A_r\,A_{r\,r}\,B^4\,C^3\,C_r - 56\,(A_r)^3\,B^4\,C^3\,C_r \\
&- 52\,A^2\,A_{t\,t}\,A_r\,B^3\,C^3\,C_r + 38\,A\,(A_t)^2\,A_r\,B^3\,C^3\,C_r + 40\,A^3\,A_{t\,t\,r}\,B^3\,C^3\,C_r \\
&- 36\,A^2\,A_t\,A_{t\,r}\,B^3\,C^3\,C_r - 8\,A^5\,B^2\,C^2\,C_t\,C_{t\,t\,t} + 8\,A^4\,A_t\,B^2\,C^3\,C_{t\,t\,t} \\
&- 16\,A^4\,B^3\,C^3\,C_{t\,t\,r\,r} + 32\,A^4\,B^2\,B_r\,C^3\,C_{t\,t\,r} + 8\,A^3\,A_r\,B^3\,C^3\,C_{t\,t\,r} \\
&+ 4\,A^5\,B^2\,C^2\,(C_{t\,t})^2 + 8\,A^5\,B^2\,C\,(C_t)^2\,C_{t\,t} + 8\,A^5\,B\,B_t\,C^2\,C_t\,C_{t\,t} \\
&- 16\,A^4\,A_t\,B^2\,C^2\,C_t\,C_{t\,t} + 24\,A^4\,B^2\,B_{r\,r}\,C^3\,C_{t\,t} - 36\,A^4\,B\,(B_r)^2\,C^3\,C_{t\,t} \\
&- 12\,A^3\,A_r\,B^2\,B_r\,C^3\,C_{t\,t} - 8\,A^4\,A_t\,B\,B_t\,C^3\,C_{t\,t} - 8\,A^4\,A_{t\,t}\,B^2\,C^3\,C_{t\,t} \\
&+ 8\,A^3\,(A_t)^2\,B^2\,C^3\,C_{t\,t} + 40\,A^4\,B^3\,C^2\,C_t\,C_{t\,r\,r} + 8\,A^4\,B^2\,B_t\,C^3\,C_{t\,r\,r}
\end{aligned}
$$

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$$
\begin{aligned}
&- 16\,A^3\,A_t\,B^3\,C^3\,C_{t\,r\,r} + 32\,A^4\,B^3\,C^2\,(C_{t\,r})^2 - 60\,A^4\,B^2\,B_r\,C^2\,C_t\,C_{t\,r} \\
&- 20\,A^3\,A_r\,B^3\,C^2\,C_t\,C_{t\,r} - 32\,A^4\,B\,B_t\,B_r\,C^3\,C_{t\,r} + 12\,A^3\,A_t\,B^2\,B_r\,C^3\,C_{t\,r} \\
&+ 16\,A^4\,B^2\,B_{t\,r}\,C^3\,C_{t\,r} - 4\,A^3\,A_r\,B^2\,B_t\,C^3\,C_{t\,r} + 24\,A^2\,A_t\,A_r\,B^3\,C^3\,C_{t\,r} \\
&- 16\,A^3\,A_{t\,r}\,B^3\,C^3\,C_{t\,r} - 4\,A^5\,B^2\,(C_t)^4 - 4\,A^5\,B\,B_t\,C\,(C_t)^3 \\
&+ 4\,A^4\,A_t\,B^2\,C\,(C_t)^3 - 32\,A^4\,B^2\,B_{r\,r}\,C^2\,(C_t)^2 + 40\,A^4\,B\,(B_r)^2\,C^2\,(C_t)^2 \\
&+ 16\,A^3\,A_r\,B^2\,B_r\,C^2\,(C_t)^2 + 4\,A^5\,B\,B_{t\,t}\,C^2\,(C_t)^2 - 7\,A^5\,(B_t)^2\,C^2\,(C_t)^2 \\
&+ 2\,A^4\,A_t\,B\,B_t\,C^2\,(C_t)^2 + 12\,A^4\,A_{t\,t}\,B^2\,C^2\,(C_t)^2 - 3\,A^3\,(A_t)^2\,B^2\,C^2\,(C_t)^2 \\
&- 12\,A^4\,B\,B_t\,B_{r\,r}\,C^3\,C_t + 28\,A^3\,A_t\,B^2\,B_{r\,r}\,C^3\,C_t + 42\,A^4\,B_t\,(B_r)^2\,C^3\,C_t \\
&- 26\,A^3\,A_t\,B\,(B_r)^2\,C^3\,C_t - 24\,A^4\,B\,B_{t\,r}\,B_r\,C^3\,C_t + 6\,A^3\,A_r\,B\,B_t\,B_r\,C^3\,C_t \\
&- 34\,A^2\,A_t\,A_r\,B^2\,B_r\,C^3\,C_t + 20\,A^3\,A_{t\,r}\,B^2\,B_r\,C^3\,C_t - 8\,A^4\,A_t\,B\,B_{t\,t}\,C^3\,C_t \\
&+ 14\,A^4\,A_t\,(B_t)^2\,C^3\,C_t - 8\,A^4\,A_{t\,t}\,B\,B_t\,C^3\,C_t + 8\,A^3\,(A_t)^2\,B\,B_t\,C^3\,C_t \\
&+ 8\,A^2\,A_t\,A_{r\,r}\,B^3\,C^3\,C_t - 20\,A\,A_t\,(A_r)^2\,B^3\,C^3\,C_t + 20\,A^2\,A_{t\,r}\,A_r\,B^3\,C^3\,C_t \\
&- 8\,A^3\,A_{t\,r\,r}\,B^3\,C^3\,C_t + 8\,A^4\,A_{t\,t\,t}\,B^2\,C^3\,C_t - 24\,A^3\,A_t\,A_{t\,t}\,B^2\,C^3\,C_t \\
&+ 10\,A^2\,(A_t)^3\,B^2\,C^3\,C_t - 16\,A^3\,B^3\,B_{r\,r\,r\,r}\,C^4 + 48\,A^3\,B^2\,B_r\,B_{r\,r\,r}\,C^4 \\
&+ 48\,A^2\,A_r\,B^3\,B_{r\,r\,r}\,C^4 + 36\,A^3\,B^2\,(B_{r\,r})^2\,C^4 - 116\,A^3\,B\,(B_r)^2\,B_{r\,r}\,C^4 \\
&- 108\,A^2\,A_r\,B^2\,B_r\,B_{r\,r}\,C^4 + 12\,A^3\,A_t\,B\,B_t\,B_{r\,r}\,C^4 + 32\,A^2\,A_{r\,r}\,B^3\,B_{r\,r}\,C^4 \\
&- 76\,A\,(A_r)^2\,B^3\,B_{r\,r}\,C^4 - 24\,A^3\,A_{t\,t}\,B^2\,B_{r\,r}\,C^4 + 4\,A^2\,(A_t)^2\,B^2\,B_{r\,r}\,C^4 \\
&+ 49\,A^3\,(B_r)^4\,C^4 + 58\,A^2\,A_r\,B\,(B_r)^3\,C^4 - 42\,A^3\,A_t\,B_t\,(B_r)^2\,C^4 \\
&- 24\,A^2\,A_{r\,r}\,B^2\,(B_r)^2\,C^4 + 57\,A\,(A_r)^2\,B^2\,(B_r)^2\,C^4 + 36\,A^3\,A_{t\,t}\,B\,(B_r)^2\,C^4 \\
&- 14\,A^2\,(A_t)^2\,B\,(B_r)^2\,C^4 + 24\,A^3\,A_t\,B\,B_{t\,r}\,B_r\,C^4 - 38\,A^2\,A_t\,A_r\,B\,B_t\,B_r\,C^4 \\
&+ 32\,A^3\,A_{t\,r}\,B\,B_t\,B_r\,C^4 + 8\,A^2\,A_{r\,r\,r}\,B^3\,B_r\,C^4 - 52\,A\,A_r\,A_{r\,r}\,B^3\,B_r\,C^4 \\
&+ 56\,(A_r)^3\,B^3\,B_r\,C^4 + 44\,A^2\,A_{t\,t}\,A_r\,B^2\,B_r\,C^4 - 30\,A\,(A_t)^2\,A_r\,B^2\,B_r\,C^4 \\
&- 32\,A^3\,A_{t\,t\,r}\,B^2\,B_r\,C^4 + 28\,A^2\,A_t\,A_{t\,r}\,B^2\,B_r\,C^4 + 4\,A^3\,(A_t)^2\,B\,B_{t\,t}\,C^4
\end{aligned}
$$

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$$+ 16 A^2 A_t A_r B^2 B_{t\,r} C^4 - 16 A^3 A_{t\,r} B^2 B_{t\,r} C^4 - 7 A^3 (A_t)^2 (B_t)^2 C^4$$

$$+ 8 A^2 A_t A_{r\,r} B^2 B_t C^4 - 20 A A_t (A_r)^2 B^2 B_t C^4 + 20 A^2 A_{t\,r} A_r B^2 B_t C^4$$

$$- 8 A^3 A_{t\,r\,r} B^2 B_t C^4 + 8 A^3 A_t A_{t\,t} B B_t C^4 - 6 A^2 (A_t)^3 B B_t C^4$$

$$- 16 A^2 A_{t\,t} A_{r\,r} B^3 C^4 + 16 A (A_t)^2 A_{r\,r} B^3 C^4 + 40 A A_{t\,t} (A_r)^2 B^3 C^4$$

$$- 56 (A_t)^2 (A_r)^2 B^3 C^4 - 40 A^2 A_{t\,t\,r} A_r B^3 C^4 + 72 A A_t A_{t\,r} A_r B^3 C^4$$

$$+ 16 A^3 A_{t\,t\,r\,r} B^3 C^4 - 16 A^2 A_t A_{t\,r\,r} B^3 C^4 - 16 A^2 (A_{t\,r})^2 B^3 C^4$$

$$- 8 A^3 A_t A_{t\,t\,t} B^2 C^4 + 4 A^3 (A_{t\,t})^2 B^2 C^4 + 12 A^2 (A_t)^2 A_{t\,t} B^2 C^4$$

$$- 7 A (A_t)^4 B^2 C^4 - 16 A^5 B^4 C^2)/(48 A^5 B^5 C^4)$$

APPENDIX B

In this appendix we present expressions for the relevant components of the tensor $W^{\mu\nu}$ in several specific coordinate systems of interest.

1. STANDARD COORDINATES

In this case the line element is given by

$$ds^2 = -B(r, t)dt^2 + A(r, t)dr^2 + r^2 d\Omega ,$$

and the relevant components of $W^{\mu\nu}$ are denoted by S^{rr}, S^{rt}, and S^{tt}. They are found to take the forms which follow below.

$$S^{rr} =$$

$$(8 A^2 B^3 B_r B_{r\,r\,r} r^4 - 4 A^2 B^3 (B_{r\,r})^2 r^4 - 12 A^2 B^2 (B_r)^2 B_{r\,r} r^4$$

$$- 8 A A_r B^3 B_r B_{r\,r} r^4 + 16 A^3 B^2 B_{t\,t} B_{r\,r} r^4 - 40 A^3 B (B_t)^2 B_{r\,r} r^4$$

$$- 44 A^2 A_t B^2 B_t B_{r\,r} r^4 + 24 A^2 A_{t\,t} B^3 B_{r\,r} r^4 - 36 A (A_t)^2 B^3 B_{r\,r} r^4$$

$$+ 7 A^2 B (B_r)^4 r^4 + 6 A A_r B^2 (B_r)^3 r^4 - 16 A^3 B B_{t\,t} (B_r)^2 r^4$$

$$+ 56 A^3 (B_t)^2 (B_r)^2 r^4 + 30 A^2 A_t B B_t (B_r)^2 r^4 - 4 A A_{r\,r} B^3 (B_r)^2 r^4$$

$$+ 7 (A_r)^2 B^3 (B_r)^2 r^4 - 4 A^2 A_{t\,t} B^2 (B_r)^2 r^4 + 14 A (A_t)^2 B^2 (B_r)^2 r^4$$

$$+ 16 A^3 B^2 B_{t\,t\,r} B_r r^4 - 8 A^2 A_r B^2 B_{t\,t} B_r r^4 - 72 A^3 B B_t B_{t\,r} B_r r^4$$

$$- 28 A^2 A_t B^2 B_{t\,r} B_r r^4 + 20 A^2 A_r B (B_t)^2 B_r r^4 + 38 A A_t A_r B^2 B_t B_r r^4$$

$$- 16 A^2 A_{t\,r} B^2 B_t B_r r^4 - 12 A A_{t\,t} A_r B^3 B_r r^4 + 42 (A_t)^2 A_r B^3 B_r r^4$$

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$$- 24\,A\,A_t\,A_{t\,r}\,B^3\,B_r\,r^4 - 8\,A^3\,A_t\,B^2\,B_{t\,t\,t}\,r^4 - 16\,A^3\,B^3\,B_{t\,t\,r\,r}\,r^4$$

$$+ 8\,A^2\,A_r\,B^3\,B_{t\,t\,r}\,r^4 + 52\,A^3\,A_t\,B\,B_t\,B_{t\,t}\,r^4 - 32\,A^3\,A_{t\,t}\,B^2\,B_{t\,t}\,r^4$$

$$+ 24\,A^2\,(A_t)^2\,B^2\,B_{t\,t}\,r^4 + 40\,A^3\,B^2\,B_t\,B_{t\,r\,r}\,r^4 + 32\,A^2\,A_t\,B^3\,B_{t\,r\,r}\,r^4$$

$$+ 16\,A^3\,B^2\,(B_{t\,r})^2\,r^4 - 20\,A^2\,A_r\,B^2\,B_t\,B_{t\,r}\,r^4 - 32\,A\,A_t\,A_r\,B^3\,B_{t\,r}\,r^4$$

$$+ 16\,A^2\,A_{t\,r}\,B^3\,B_{t\,r}\,r^4 - 56\,A^3\,A_t\,(B_t)^3\,r^4 + 76\,A^3\,A_{t\,t}\,B\,(B_t)^2\,r^4$$

$$- 57\,A^2\,(A_t)^2\,B\,(B_t)^2\,r^4 - 48\,A^3\,A_{t\,t\,t}\,B^2\,B_t\,r^4 + 108\,A^2\,A_t\,A_{t\,t}\,B^2\,B_t\,r^4$$

$$- 58\,A\,(A_t)^3\,B^2\,B_t\,r^4 + 16\,A^3\,A_{t\,t\,t\,t}\,B^3\,r^4 - 48\,A^2\,A_t\,A_{t\,t\,t}\,B^3\,r^4$$

$$- 36\,A^2\,(A_{t\,t})^2\,B^3\,r^4 + 116\,A\,(A_t)^2\,A_{t\,t}\,B^3\,r^4 - 49\,(A_t)^4\,B^3\,r^4$$

$$- 16\,A^2\,B^4\,B_{r\,r\,r}\,r^3 + 48\,A^2\,B^3\,B_r\,B_{r\,r}\,r^3 + 16\,A\,A_r\,B^4\,B_{r\,r}\,r^3$$

$$- 20\,A^2\,B^2\,(B_r)^3\,r^3 - 16\,A\,A_r\,B^3\,(B_r)^2\,r^3 - 16\,A^3\,B^2\,B_{t\,t}\,B_r\,r^3$$

$$+ 40\,A^3\,B\,(B_t)^2\,B_r\,r^3 + 68\,A^2\,A_t\,B^2\,B_t\,B_r\,r^3 + 16\,A\,A_{r\,r}\,B^4\,B_r\,r^3$$

$$- 28\,(A_r)^2\,B^4\,B_r\,r^3 - 56\,A^2\,A_{t\,t}\,B^3\,B_r\,r^3 + 52\,A\,(A_t)^2\,B^3\,B_r\,r^3$$

$$+ 16\,A^3\,B^3\,B_{t\,t\,r}\,r^3 - 40\,A^3\,B^2\,B_t\,B_{t\,r}\,r^3 - 40\,A^2\,A_t\,B^3\,B_{t\,r}\,r^3$$

$$- 12\,A\,A_t\,A_r\,B^3\,B_t\,r^3 + 24\,A\,A_{t\,t}\,A_r\,B^4\,r^3 - 84\,(A_t)^2\,A_r\,B^4\,r^3$$

$$+ 48\,A\,A_t\,A_{t\,r}\,B^4\,r^3 - 32\,A^2\,B^4\,B_{r\,r}\,r^2 - 4\,A^2\,B^3\,(B_r)^2\,r^2 + 8\,A\,A_r\,B^4\,B_r\,r^2$$

$$- 40\,A^2\,A_t\,B^3\,B_t\,r^2 - 16\,A\,A_{r\,r}\,B^5\,r^2 + 28\,(A_r)^2\,B^5\,r^2 + 80\,A^2\,A_{t\,t}\,B^4\,r^2$$

$$- 88\,A\,(A_t)^2\,B^4\,r^2 + 32\,A^2\,B^4\,B_r\,r + 16\,A^4\,B^5 - 16\,A^2\,B^5)/(48\,A^5\,B^5\,r^4)$$

$$s^{rt} =$$

$$- (4\,A^3\,B^2\,B_t\,B_{r\,r\,r}\,r^3 + 6\,A^2\,A_t\,B^3\,B_{r\,r\,r}\,r^3 - 18\,A^3\,B\,B_t\,B_r\,B_{r\,r}\,r^3$$

$$- 14\,A^2\,A_t\,B^2\,B_r\,B_{r\,r}\,r^3 + 8\,A^3\,B^2\,B_{t\,r}\,B_{r\,r}\,r^3 - 6\,A^2\,A_r\,B^2\,B_t\,B_{r\,r}\,r^3$$

$$- 15\,A\,A_t\,A_r\,B^3\,B_{r\,r}\,r^3 + 6\,A^2\,A_{t\,r}\,B^3\,B_{r\,r}\,r^3 + 14\,A^3\,B_t\,(B_r)^3\,r^3$$

$$+ 7\,A^2\,A_t\,B\,(B_r)^3\,r^3 - 14\,A^3\,B\,B_{t\,r}\,(B_r)^2\,r^3 + 9\,A^2\,A_r\,B\,B_t\,(B_r)^2\,r^3$$

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$$+ 12\,A\,A_t\,A_r\,B^2\,(B_r)^2\,r^3 - 5\,A^2\,A_{t\,r}\,B^2\,(B_r)^2\,r^3 + 5\,A^3\,A_t\,B\,B_{t\,t}\,B_r\,r^3$$

$$+ 10\,A^3\,B^2\,B_{t\,r\,r}\,B_r\,r^3 - 9\,A^2\,A_r\,B^2\,B_{t\,r}\,B_r\,r^3 - 14\,A^3\,A_t\,(B_t)^2\,B_r\,r^3$$

$$- 2\,A^2\,A_{r\,r}\,B^2\,B_t\,B_r\,r^3 + 4\,A\,(A_r)^2\,B^2\,B_t\,B_r\,r^3 + 15\,A^3\,A_{t\,t}\,B\,B_t\,B_r\,r^3$$

$$- 12\,A^2\,(A_t)^2\,B\,B_t\,B_r\,r^3 - 5\,A\,A_t\,A_{r\,r}\,B^3\,B_r\,r^3 + 14\,A_t\,(A_r)^2\,B^3\,B_r\,r^3$$

$$- 8\,A\,A_{t\,r}\,A_r\,B^3\,B_r\,r^3 + 2\,A^2\,A_{t\,r\,r}\,B^3\,B_r\,r^3 - 6\,A^3\,A_{t\,t\,t}\,B^2\,B_r\,r^3$$

$$+ 14\,A^2\,A_t\,A_{t\,t}\,B^2\,B_r\,r^3 - 7\,A\,(A_t)^3\,B^2\,B_r\,r^3 - 2\,A^3\,A_t\,B^2\,B_{t\,t\,r}\,r^3$$

$$+ 2\,A^2\,A_t\,A_r\,B^2\,B_{t\,t}\,r^3 - 2\,A^3\,A_{t\,r}\,B^2\,B_{t\,t}\,r^3 - 4\,A^3\,B^3\,B_{t\,r\,r\,r}\,r^3$$

$$+ 6\,A^2\,A_r\,B^3\,B_{t\,r\,r}\,r^3 + 8\,A^3\,A_t\,B\,B_t\,B_{t\,r}\,r^3 + 2\,A^2\,A_{r\,r}\,B^3\,B_{t\,r}\,r^3$$

$$- 4\,A\,(A_r)^2\,B^3\,B_{t\,r}\,r^3 - 6\,A^3\,A_{t\,t}\,B^2\,B_{t\,r}\,r^3 + 5\,A^2\,(A_t)^2\,B^2\,B_{t\,r}\,r^3$$

$$- 4\,A^2\,A_t\,A_r\,B\,(B_t)^2\,r^3 + 4\,A^3\,A_{t\,r}\,B\,(B_t)^2\,r^3 + 6\,A^2\,A_{t\,t}\,A_r\,B^2\,B_t\,r^3$$

$$- 9\,A\,(A_t)^2\,A_r\,B^2\,B_t\,r^3 - 6\,A^3\,A_{t\,t\,r}\,B^2\,B_t\,r^3 + 9\,A^2\,A_t\,A_{t\,r}\,B^2\,B_t\,r^3$$

$$- 4\,A^2\,A_{t\,t\,t}\,A_r\,B^3\,r^3 + 18\,A\,A_t\,A_{t\,t}\,A_r\,B^3\,r^3 - 14\,(A_t)^3\,A_r\,B^3\,r^3$$

$$+ 4\,A^3\,A_{t\,t\,t\,r}\,B^3\,r^3 - 10\,A^2\,A_t\,A_{t\,t\,r}\,B^3\,r^3 - 8\,A^2\,A_{t\,r}\,A_{t\,t}\,B^3\,r^3$$

$$+ 14\,A\,(A_t)^2\,A_{t\,r}\,B^3\,r^3 + 8\,A^3\,B^2\,B_t\,B_{r\,r}\,r^2 + 10\,A^2\,A_t\,B^3\,B_{r\,r}\,r^2$$

$$- 2\,A^3\,B\,B_t\,(B_r)^2\,r^2 + 2\,A^3\,B^2\,B_{t\,r}\,B_r\,r^2 - 2\,A^2\,A_r\,B^2\,B_t\,B_r\,r^2$$

$$- 8\,A\,A_t\,A_r\,B^3\,B_r\,r^2 + 4\,A^2\,A_{t\,r}\,B^3\,B_r\,r^2 - 6\,A^3\,A_t\,B^2\,B_{t\,t}\,r^2$$

$$- 8\,A^3\,B^3\,B_{t\,r\,r}\,r^2 + 2\,A^2\,A_r\,B^3\,B_{t\,r}\,r^2 + 12\,A^3\,A_t\,B\,(B_t)^2\,r^2$$

$$- 18\,A^3\,A_{t\,t}\,B^2\,B_t\,r^2 + 14\,A^2\,(A_t)^2\,B^2\,B_t\,r^2 + 10\,A\,A_t\,A_{r\,r}\,B^4\,r^2$$

$$- 28\,A_t\,(A_r)^2\,B^4\,r^2 + 16\,A\,A_{t\,r}\,A_r\,B^4\,r^2 - 4\,A^2\,A_{t\,r\,r}\,B^4\,r^2$$

$$+ 12\,A^3\,A_{t\,t\,t}\,B^3\,r^2 - 28\,A^2\,A_t\,A_{t\,t}\,B^3\,r^2 + 14\,A\,(A_t)^3\,B^3\,r^2 - 8\,A^3\,B^2\,B_t\,B_r\,r$$

$$- 14\,A^2\,A_t\,B^3\,B_r\,r + 8\,A^3\,B^3\,B_{t\,r}\,r - 2\,A\,A_t\,A_r\,B^4\,r + 8\,A^2\,A_t\,B^4)$$

$$/(12\,A^5\,B^5\,r^3)$$

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$$
\begin{aligned}
S^{tt} = & \\
& - (16 A^3 B^3 B_{r\,r\,r\,r} r^4 - 48 A^3 B^2 B_r B_{r\,r\,r} r^4 - 48 A^2 A_r B^3 B_{r\,r\,r} r^4 \\
& - 36 A^3 B^2 (B_{r\,r})^2 r^4 + 116 A^3 B (B_r)^2 B_{r\,r} r^4 + 108 A^2 A_r B^2 B_r B_{r\,r} r^4 \\
& - 12 A^3 A_t B B_t B_{r\,r} r^4 - 32 A^2 A_{r\,r} B^3 B_{r\,r} r^4 + 76 A (A_r)^2 B^3 B_{r\,r} r^4 \\
& + 24 A^3 A_{t\,t} B^2 B_{r\,r} r^4 - 4 A^2 (A_t)^2 B^2 B_{r\,r} r^4 - 49 A^3 (B_r)^4 r^4 \\
& - 58 A^2 A_r B (B_r)^3 r^4 + 42 A^3 A_t B_t (B_r)^2 r^4 + 24 A^2 A_{r\,r} B^2 (B_r)^2 r^4 \\
& - 57 A (A_r)^2 B^2 (B_r)^2 r^4 - 36 A^3 A_{t\,t} B (B_r)^2 r^4 + 14 A^2 (A_t)^2 B (B_r)^2 r^4 \\
& - 24 A^3 A_t B B_{t\,r} B_r r^4 + 38 A^2 A_t A_r B B_t B_r r^4 - 32 A^3 A_{t\,r} B B_t B_r r^4 \\
& - 8 A^2 A_{r\,r\,r} B^3 B_r r^4 + 52 A A_r A_{r\,r} B^3 B_r r^4 - 56 (A_r)^3 B^3 B_r r^4 \\
& - 44 A^2 A_{t\,t} A_r B^2 B_r r^4 + 30 A (A_t)^2 A_r B^2 B_r r^4 + 32 A^3 A_{t\,t\,r} B^2 B_r r^4 \\
& - 28 A^2 A_t A_{t\,r} B^2 B_r r^4 - 4 A^3 (A_t)^2 B B_{t\,t} r^4 - 16 A^2 A_t A_r B^2 B_{t\,r} r^4 \\
& + 16 A^3 A_{t\,r} B^2 B_{t\,r} r^4 + 7 A^3 (A_t)^2 (B_t)^2 r^4 - 8 A^2 A_t A_{r\,r} B^2 B_t r^4 \\
& + 20 A A_t (A_r)^2 B^2 B_t r^4 - 20 A^2 A_{t\,r} A_r B^2 B_t r^4 + 8 A^3 A_{t\,r\,r} B^2 B_t r^4 \\
& - 8 A^3 A_t A_{t\,t} B B_t r^4 + 6 A^2 (A_t)^3 B B_t r^4 + 16 A^2 A_{t\,t} A_{r\,r} B^3 r^4 \\
& - 16 A (A_t)^2 A_{r\,r} B^3 r^4 - 40 A A_{t\,t} (A_r)^2 B^3 r^4 + 56 (A_t)^2 (A_r)^2 B^3 r^4 \\
& + 40 A^2 A_{t\,t\,r} A_r B^3 r^4 - 72 A A_t A_{t\,r} A_r B^3 r^4 - 16 A^3 A_{t\,t\,r\,r} B^3 r^4 \\
& + 16 A^2 A_t A_{t\,r\,r} B^3 r^4 + 16 A^2 (A_{t\,r})^2 B^3 r^4 + 8 A^3 A_t A_{t\,t\,t} B^2 r^4 \\
& - 4 A^3 (A_{t\,t})^2 B^2 r^4 - 12 A^2 (A_t)^2 A_{t\,t} B^2 r^4 + 7 A (A_t)^4 B^2 r^4 \\
& + 64 A^3 B^3 B_{r\,r\,r} r^3 - 104 A^3 B^2 B_r B_{r\,r} r^3 - 104 A^2 A_r B^3 B_{r\,r} r^3 \\
& + 44 A^3 B (B_r)^3 r^3 + 48 A^2 A_r B^2 (B_r)^2 r^3 - 84 A^3 A_t B B_t B_r r^3 \\
& - 24 A^2 A_{r\,r} B^3 B_r r^3 + 52 A (A_r)^2 B^3 B_r r^3 + 72 A^3 A_{t\,t} B^2 B_r r^3 \\
& - 44 A^2 (A_t)^2 B^2 B_r r^3 + 48 A^3 A_t B^2 B_{t\,r} r^3 - 52 A^2 A_t A_r B^2 B_t r^3 \\
& + 40 A^3 A_{t\,r} B^2 B_t r^3 + 16 A^2 A_{r\,r\,r} B^4 r^3 - 104 A A_r A_{r\,r} B^4 r^3
\end{aligned}
$$

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$$+ 112 (A_r)^3 B^4 r^3 + 104 A^2 A_{t\,t} A_r B^3 r^3 - 76 A (A_t)^2 A_r B^3 r^3$$

$$- 80 A^3 A_{t\,t\,r} B^3 r^3 + 72 A^2 A_t A_{t\,r} B^3 r^3 + 20 A^3 B^2 (B_r)^2 r^2$$

$$+ 40 A^2 A_r B^3 B_r r^2 + 24 A^3 A_t B^2 B_t r^2 + 16 A^2 A_{r\,r} B^4 r^2 - 28 A (A_r)^2 B^4 r^2$$

$$- 48 A^3 A_{t\,t} B^3 r^2 + 40 A^2 (A_t)^2 B^3 r^2 - 32 A^2 A_r B^4 r + 16 A^5 B^4 - 16 A^3 B^4)$$

$$/(48 A^5 B^5 r^4)$$

2. STATIC STANDARD COORDINATES

In this case the line element is given by

$$ds^2 = -B(r)dt^2 + A(r)dr^2 + r^2 d\Omega ,$$

and the relevant components of $W^{\mu\nu}$ are denoted by SS^{rr} and SS^{tt} (in this case SS^{rt} vanishes identically). The forms which follow below are the ones which we used in Paper I.

$$SS^{rr} =$$

$$(8 A^2 B^2 B_r B_{r\,r\,r} r^4 - 4 A^2 B^2 (B_{r\,r})^2 r^4 - 12 A^2 B (B_r)^2 B_{r\,r} r^4$$

$$- 8 A A_r B^2 B_r B_{r\,r} r^4 + 7 A^2 (B_r)^4 r^4 + 6 A A_r B (B_r)^3 r^4$$

$$- 4 A A_{r\,r} B^2 (B_r)^2 r^4 + 7 (A_r)^2 B^2 (B_r)^2 r^4 - 16 A^2 B^3 B_{r\,r\,r} r^3$$

$$+ 48 A^2 B^2 B_r B_{r\,r} r^3 + 16 A A_r B^3 B_{r\,r} r^3 - 20 A^2 B (B_r)^3 r^3$$

$$- 16 A A_r B^2 (B_r)^2 r^3 + 16 A A_{r\,r} B^3 B_r r^3 - 28 (A_r)^2 B^3 B_r r^3$$

$$- 32 A^2 B^3 B_{r\,r} r^2 - 4 A^2 B^2 (B_r)^2 r^2 + 8 A A_r B^3 B_r r^2 - 16 A A_{r\,r} B^4 r^2$$

$$+ 28 (A_r)^2 B^4 r^2 + 32 A^2 B^3 B_r r + 16 A^4 B^4 - 16 A^2 B^4)/(48 A^5 B^4 r^4)$$

$$SS^{tt} =$$

$$- (16 A^3 B^3 B_{r\,r\,r\,r} r^4 - 48 A^3 B^2 B_r B_{r\,r\,r} r^4 - 48 A^2 A_r B^3 B_{r\,r\,r} r^4$$

$$- 36 A^3 B^2 (B_{r\,r})^2 r^4 + 116 A^3 B (B_r)^2 B_{r\,r} r^4 + 108 A^2 A_r B^2 B_r B_{r\,r} r^4$$

$$- 32 A^2 A_{r\,r} B^3 B_{r\,r} r^4 + 76 A (A_r)^2 B^3 B_{r\,r} r^4 - 49 A^3 (B_r)^4 r^4$$

$$- 58 A^2 A_r B (B_r)^3 r^4 + 24 A^2 A_{r\,r} B^2 (B_r)^2 r^4 - 57 A (A_r)^2 B^2 (B_r)^2 r^4$$

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$$- 8 A^2 A_{r\,r\,r} B^3 B_r r^4 + 52 A A_r A_{r\,r} B^3 B_r r^4 - 56 (A_r)^3 B^3 B_r r^4$$

$$+ 64 A^3 B^3 B_{r\,r\,r} r^3 - 104 A^3 B^2 B_r B_{r\,r} r^3 - 104 A^2 A_r B^3 B_{r\,r} r^3$$

$$+ 44 A^3 B (B_r)^3 r^3 + 48 A^2 A_r B^2 (B_r)^2 r^3 - 24 A^2 A_{r\,r} B^3 B_r r^3$$

$$+ 52 A (A_r)^2 B^3 B_r r^3 + 16 A^2 A_{r\,r\,r} B^4 r^3 - 104 A A_r A_{r\,r} B^4 r^3$$

$$+ 112 (A_r)^3 B^4 r^3 + 20 A^3 B^2 (B_r)^2 r^2 + 40 A^2 A_r B^3 B_r r^2 + 16 A^2 A_{r\,r} B^4 r^2$$

$$- 28 A (A_r)^2 B^4 r^2 - 32 A^2 A_r B^4 r + 16 A^5 B^4 - 16 A^3 B^4)/(48 A^5 B^5 r^4)$$

3. ISOTROPIC COORDINATES

In this case the line element is given by

$$ds^2 = -B(r, t)dt^2 + A(r, t)(dr^2 + r^2 d\Omega)$$

and the relevant components of $W^{\mu\nu}$ are denoted by I^{rr}, I^{rt}, and I^{tt}. They are found to take the forms which follow below.

$$I^{rr} =$$

$$(8 A^4 B^3 B_r B_{r\,r\,r} r^3 - 8 A^3 A_r B^4 B_{r\,r\,r} r^3 - 4 A^4 B^3 (B_{r\,r})^2 r^3$$

$$- 12 A^4 B^2 (B_r)^2 B_{r\,r} r^3 + 16 A^3 A_r B^3 B_r B_{r\,r} r^3 + 16 A^5 B^2 B_{t\,t} B_{r\,r} r^3$$

$$- 40 A^5 B (B_t)^2 B_{r\,r} r^3 + 8 A^4 A_t B^2 B_t B_{r\,r} r^3 + 8 A^3 A_{r\,r} B^4 B_{r\,r} r^3$$

$$- 4 A^2 (A_r)^2 B^4 B_{r\,r} r^3 + 7 A^4 B (B_r)^4 r^3 - 4 A^3 A_r B^2 (B_r)^3 r^3$$

$$- 16 A^5 B B_{t\,t} (B_r)^2 r^3 + 56 A^5 (B_t)^2 (B_r)^2 r^3 - 8 A^4 A_t B B_t (B_r)^2 r^3$$

$$- 12 A^3 A_{r\,r} B^3 (B_r)^2 r^3 + 2 A^2 (A_r)^2 B^3 (B_r)^2 r^3 + 16 A^5 B^2 B_{t\,t\,r} B_r r^3$$

$$- 16 A^4 A_r B^2 B_{t\,t} B_r r^3 - 72 A^5 B B_t B_{t\,r} B_r r^3 + 8 A^4 A_t B^2 B_{t\,r} B_r r^3$$

$$+ 40 A^4 A_r B (B_t)^2 B_r r^3 + 32 A^3 A_t A_r B^2 B_t B_r r^3 - 40 A^4 A_{t\,r} B^2 B_t B_r r^3$$

$$- 8 A^3 A_{r\,r\,r} B^4 B_r r^3 + 32 A^2 A_r A_{r\,r} B^4 B_r r^3 - 20 A (A_r)^3 B^4 B_r r^3$$

$$- 16 A^3 A_{t\,t} A_r B^3 B_r r^3 + 24 A^2 (A_t)^2 A_r B^3 B_r r^3 + 16 A^4 A_{t\,t\,r} B^3 B_r r^3$$

$$- 24 A^3 A_t A_{t\,r} B^3 B_r r^3 - 16 A^5 B^3 B_{t\,t\,r\,r} r^3 + 16 A^4 A_r B^3 B_{t\,t\,r} r^3$$

$$+ 40 A^5 B^2 B_t B_{t\,r\,r} r^3 - 8 A^4 A_t B^3 B_{t\,r\,r} r^3 + 16 A^5 B^2 (B_{t\,r})^2 r^3$$

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$$- 40 A^4 A_r B^2 B_t B_{t\,r} r^3 - 24 A^3 A_t A_r B^3 B_{t\,r} r^3 + 32 A^4 A_{t\,r} B^3 B_{t\,r} r^3$$
$$+ 8 A^3 A_t A_{r\,r} B^3 B_t r^3 - 24 A^2 A_t (A_r)^2 B^3 B_t r^3 + 24 A^3 A_{t\,r} A_r B^3 B_t r^3$$
$$- 8 A^4 A_{t\,r\,r} B^3 B_t r^3 + 8 A^2 A_r A_{r\,r} B^5 r^3 - 4 A^2 (A_{r\,r})^2 B^5 r^3$$
$$- 20 A (A_r)^2 A_{r\,r} B^5 r^3 + 15 (A_r)^4 B^5 r^3 - 16 A^3 A_{t\,t} A_{r\,r} B^4 r^3$$
$$+ 24 A^2 (A_t)^2 A_{r\,r} B^4 r^3 + 48 A^2 A_{t\,t} (A_r)^2 B^4 r^3 - 120 A (A_t)^2 (A_r)^2 B^4 r^3$$
$$- 48 A^3 A_{t\,t\,r} A_r B^4 r^3 + 168 A^2 A_t A_{t\,r} A_r B^4 r^3 + 16 A^4 A_{t\,t\,r\,r} B^4 r^3$$
$$- 24 A^3 A_t A_{t\,r\,r} B^4 r^3 - 48 A^3 (A_{t\,r})^2 B^4 r^3 - 16 A^4 B^4 B_{r\,r\,r} r^2$$
$$+ 48 A^4 B^3 B_r B_{r\,r} r^2 - 20 A^4 B^2 (B_r)^3 r^2 - 36 A^3 A_r B^3 (B_r)^2 r^2$$
$$- 16 A^5 B^2 B_{t\,t} B_r r^2 + 40 A^5 B (B_t)^2 B_r r^2 - 8 A^4 A_t B^2 B_t B_r r^2$$
$$+ 36 A^2 (A_r)^2 B^4 B_r r^2 + 16 A^5 B^3 B_{t\,t\,r} r^2 - 40 A^5 B^2 B_t B_{t\,r} r^2$$
$$+ 8 A^4 A_t B^3 B_{t\,r} r^2 - 8 A^3 A_t A_r B^3 B_t r^2 + 8 A^4 A_{t\,r} B^3 B_t r^2$$
$$+ 16 A^3 A_{r\,r\,r} B^5 r^2 - 48 A^2 A_r A_{r\,r} B^5 r^2 + 20 A (A_r)^3 B^5 r^2$$
$$+ 16 A^3 A_{t\,t} A_r B^4 r^2 - 24 A^2 (A_t)^2 A_r B^4 r^2 - 16 A^4 A_{t\,t\,r} B^4 r^2$$
$$+ 24 A^3 A_t A_{t\,r} B^4 r^2 - 32 A^4 B^4 B_{r\,r} r - 4 A^4 B^3 (B_r)^2 r + 72 A^3 A_r B^4 B_r r$$
$$+ 32 A^3 A_{r\,r} B^5 r - 68 A^2 (A_r)^2 B^5 r + 32 A^4 B^4 B_r - 32 A^3 A_r B^5)/(48 A^7 B^5 r^3)$$

$$I^{rt} =$$

$$- (2 A^4 B^2 B_t B_{r\,r\,r} r^2 - 9 A^4 B B_t B_r B_{r\,r} r^2 + 4 A^4 B^2 B_{t\,r} B_{r\,r} r^2$$
$$- A^3 A_r B^2 B_t B_{r\,r} r^2 - 2 A^2 A_t A_r B^3 B_{r\,r} r^2 + 2 A^3 A_{t\,r} B^3 B_{r\,r} r^2$$
$$+ 7 A^4 B_t (B_r)^3 r^2 - 7 A^4 B B_{t\,r} (B_r)^2 r^2 + 4 A^3 A_r B B_t (B_r)^2 r^2$$
$$+ 3 A^2 A_t A_r B^2 (B_r)^2 r^2 - 3 A^3 A_{t\,r} B^2 (B_r)^2 r^2 + 5 A^4 B^2 B_{t\,r\,r} B_r r^2$$
$$- 4 A^3 A_r B^2 B_{t\,r} B_r r^2 - 2 A^3 A_{r\,r} B^2 B_t B_r r^2 + A^2 (A_r)^2 B^2 B_t B_r r^2$$

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$$- A^2 A_t A_{r\,r} B^3 B_r r^2 + A^3 A_{t\,r\,r} B^3 B_r r^2 - 2 A^4 B^3 B_{t\,r\,r\,r} r^2$$
$$+ A^3 A_r B^3 B_{t\,r\,r} r^2 + 2 A^3 A_{r\,r} B^3 B_{t\,r} r^2 - A^2 (A_r)^2 B^3 B_{t\,r} r^2$$
$$- 2 A^2 A_t A_{r\,r\,r} B^4 r^2 + 15 A A_t A_r A_{r\,r} B^4 r^2 - 8 A^2 A_{t\,r} A_{r\,r} B^4 r^2$$
$$- 15 A_t (A_r)^3 B^4 r^2 + 15 A A_{t\,r} (A_r)^2 B^4 r^2 - 7 A^2 A_{t\,r\,r} A_r B^4 r^2$$
$$+ 2 A^3 A_{t\,r\,r\,r} B^4 r^2 + 4 A^4 B^2 B_t B_{r\,r} r - A^4 B B_t (B_r)^2 r + A^4 B^2 B_{t\,r} B_r r$$
$$- 7 A^3 A_r B^2 B_t B_r r - 7 A^2 A_t A_r B^3 B_r r + 7 A^3 A_{t\,r} B^3 B_r r$$
$$- 4 A^4 B^3 B_{t\,r\,r} r + 7 A^3 A_r B^3 B_{t\,r} r - 4 A^2 A_t A_{r\,r} B^4 r + 15 A A_t (A_r)^2 B^4 r$$
$$- 15 A^2 A_{t\,r} A_r B^4 r + 4 A^3 A_{t\,r\,r} B^4 r - 4 A^4 B^2 B_t B_r + 4 A^4 B^3 B_{t\,r}$$
$$+ 4 A^2 A_t A_r B^4 - 4 A^3 A_{t\,r} B^4)/(6 A^6 B^5 r^2)$$

$$I^{tt}=$$

$$- (16 A^4 B^3 B_{r\,r\,r\,r} r^2 - 48 A^4 B^2 B_r B_{r\,r\,r} r^2 - 16 A^3 A_r B^3 B_{r\,r\,r} r^2$$
$$- 36 A^4 B^2 (B_{r\,r})^2 r^2 + 116 A^4 B (B_r)^2 B_{r\,r} r^2 + 56 A^3 A_r B^2 B_r B_{r\,r} r^2$$
$$- 24 A^3 A_{r\,r} B^3 B_{r\,r} r^2 + 20 A^2 (A_r)^2 B^3 B_{r\,r} r^2 - 49 A^4 (B_r)^4 r^2$$
$$- 36 A^3 A_r B (B_r)^3 r^2 + 28 A^3 A_{r\,r} B^2 (B_r)^2 r^2 - 30 A^2 (A_r)^2 B^2 (B_r)^2 r^2$$
$$- 16 A^3 A_{r\,r\,r} B^3 B_r r^2 + 40 A^2 A_r A_{r\,r} B^3 B_r r^2 - 20 A (A_r)^3 B^3 B_r r^2$$
$$- 16 A^3 A_{r\,r\,r\,r} B^4 r^2 + 80 A^2 A_r A_{r\,r\,r} B^4 r^2 + 60 A^2 (A_{r\,r})^2 B^4 r^2$$
$$- 260 A (A_r)^2 A_{r\,r} B^4 r^2 + 135 (A_r)^4 B^4 r^2 + 64 A^4 B^3 B_{r\,r\,r} r$$
$$- 104 A^4 B^2 B_r B_{r\,r} r - 88 A^3 A_r B^3 B_{r\,r} r + 44 A^4 B (B_r)^3 r$$
$$+ 76 A^3 A_r B^2 (B_r)^2 r - 88 A^3 A_{r\,r} B^3 B_r r + 100 A^2 (A_r)^2 B^3 B_r r$$
$$- 64 A^3 A_{r\,r\,r} B^4 r + 280 A^2 A_r A_{r\,r} B^4 r - 220 A (A_r)^3 B^4 r + 20 A^4 B^2 (B_r)^2$$
$$- 40 A^3 A_r B^3 B_r + 20 A^2 (A_r)^2 B^4)/(48 A^6 B^5 r^2)$$

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4. COMOVING COORDINATES

In this case the line element is given by

$$ds^2 = -dt^2 + A(r, t)dr^2 + C(r, t)d\Omega\,,$$

and the relevant components of $W^{\mu\nu}$ are denoted by C^{rr}, C^{rt}, and C^{tt}. They are found to take the forms which follow below.

$$
\begin{aligned}
C^{rr} = {} & (8A^2C^2C_rC_{rrr} - 4A^2C^2(C_{rr})^2 - 8A^2C(C_r)^2C_{rr} \\
& - 8AA_rC^2C_rC_{rr} + 8A^3C^2C_{ttrr} - 8A^3C(C_t)^2C_{rr} \\
& - 4A^2A_tC^2C_tC_{rr} - 24A^2A_{tt}C^3C_{rr} + 36A(A_t)^2C^3C_{rr} + 4A^2(C_r)^4 \\
& + 4AA_rC(C_r)^3 + 8A^3CC_{ttr}(C_r)^2 - 24A^3(C_t)^2(C_r)^2 - 28A^2A_tCC_t(C_r)^2 \\
& - 4AA_{rr}C^2(C_r)^2 + 7(A_r)^2C^2(C_r)^2 + 32A^2A_{tt}C^2(C_r)^2 \\
& - 40A(A_t)^2C^2(C_r)^2 - 40A^3\epsilon^2C_{ttr}C_r - 4A^2A_rC^2C_{tt}C_r \\
& + 64A^3CC_tC_{ttr}C_r + 60A^2A_tC^2C_{ttr}C_r + 4A^2A_rC(C_t)^2C_r \\
& + 10AA_tA_rC^2C_tC_r - 8A^2A_{tr}C^2C_tC_r + 12AA_{tt}A_rC^3C_r \\
& - 42(A_t)^2A_rC^3C_r + 24AA_tA_{tr}C^3C_r - 16A^4C^3C_{tttt} \\
& + 24A^4C^2C_tC_{ttt} - 8A^3A_tC^3C_{ttt} + 16A^3C^3C_{ttrr} \\
& - 8A^2A_rC^3C_{ttr} + 28A^4C^2(C_{tt})^2 - 56A^4C(C_t)^2C_{tt} \\
& - 8A^3A_tC^2C_tC_{tt} + 8A^3A_{tt}C^3C_{tt} + 4A^2(A_t)^2C^3C_{tt} \\
& + 8A^3C^2C_tC_{trr} - 32A^2A_tC^3C_{trr} - 32A^3C^2(C_{tr})^2 \\
& - 4A^2A_rC^2C_tC_{tr} + 32AA_tA_rC^3C_{tr} - 16A^2A_{tr}C^3C_{tr} + 20A^4(C_t)^4 \\
& + 4A^3A_tC(C_t)^3 - 4A^3A_{tt}C^2(C_t)^2 + 3A^2(A_t)^2C^2(C_t)^2 \\
& + 32A^3A_{ttt}C^3C_t - 52A^2A_tA_{tt}C^3C_t + 22A(A_t)^3C^3C_t \\
& + 16A^3A_{tttt}C^4 - 48A^2A_tA_{ttt}C^4 - 36A^2(A_{tt})^2C^4 \\
& + 116A(A_t)^2A_{tt}C^4 - 49(A_t)^4C^4 + 16A^4C^2)/(48A^5C^4)
\end{aligned}
$$

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$$
\begin{aligned}
C^{rt} = & \\
& - (2A^3C^2C_tC_{rrr} - 6A^2A_tC^3C_{rrr} + 4A^3CC_tC_rC_{rr} \\
& + 10A^2A_tC^2C_rC_{rr} - 8A^3C^2C_{tr}C_{rr} - 3A^2A_rC^2C_tC_{rr} \\
& + 15AA_tA_rC^3C_{rr} - 6A^2A_{tr}C^3C_{rr} - 4A^3C_t(C_r)^3 - 4A^2A_tC(C_r)^3 \\
& + 8A^3CC_{tr}(C_r)^2 - 2A^2A_rCC_t(C_r)^2 - 8AA_tA_rC^2(C_r)^2 \\
& + 3A^2A_{tr}C^2(C_r)^2 - 2A^4C^2C_{ttt}C_r - 4A^4CC_tC_{tt}C_r - A^3A_tC^2C_{tt}C_r \\
& - 6A^3C^2C_{trr}C_r + 7A^2A_rC^2C_{tr}C_r + 4A^4(C_t)^3C_r + 2A^3A_tC(C_t)^2C_r \\
& - A^2A_{rr}C^2C_tC_r + 2A(A_r)^2C^2C_tC_r + 3A^3A_{tt}C^2C_tC_r \\
& - A^2(A_t)^2C^2C_tC_r + 5AA_tA_{rr}C^3C_r - 14A_t(A_r)^2C^3C_r + 8AA_{tr}A_rC^3C_r \\
& + 6A^3A_{ttt}C^3C_r - 14A^2A_tA_{tt}C^3C_r - 2A^2A_{trr}C^3C_r + 7A(A_t)^3C^3C_r \\
& - 4A^4C^3C_{ttt} + 6A^4C^2C_tC_{tt} + 8A^4C^2C_{tr}C_t + 2A^2A_tA_rC^3C_{tt} \\
& - 2A^3A_{tr}C^3C_{tt} + 4A^3C^3C_{trr} - 6A^2A_rC^3C_{trr} - 8A^4C(C_t)^2C_{tr} \\
& - 5A^3A_tC^2C_tC_{tr} - 2A^2A_{rr}C^3C_{tr} + 4A(A_r)^2C^3C_{tr} \\
& + 2A^3A_{tt}C^3C_{tr} + A^2(A_t)^2C^3C_{tr} + A^2A_tA_rC^2(C_t)^2 - A^3A_{tr}C^2(C_t)^2 \\
& - 4A^2A_{tt}A_rC^3C_t + A(A_t)^2A_rC^3C_t + 4A^3A_{ttr}C^3C_t - A^2A_tA_{tr}C^3C_t \\
& - 4A^2A_{ttt}A_rC^4 + 18AA_tA_{tt}A_rC^4 - 14(A_t)^3A_rC^4 + 4A^3A_{ttt}C^4 \\
& - 10A^2A_tA_{ttr}C^4 - 8A^2A_{tr}A_{tt}C^4 + 14A(A_t)^2A_{tr}C^4)/(12A^5C^4)
\end{aligned}
$$

$$
\begin{aligned}
C^{tt} = & \\
& (16A^3C^3C_{rrrr} - 24A^3C^2C_rC_{rrr} - 48A^2A_rC^3C_{rrr} \\
& - 28A^3C^2(C_{rr})^2 + 56A^3C(C_r)^2C_{rr} + 64A^2A_rC^2C_rC_{rr} \\
& - 8A^4C^2C_{tt}C_{rr} - 8A^4C(C_t)^2C_{rr} - 12A^3A_tC^2C_tC_{rr} \\
& - 32A^2A_{rr}C^3C_{rr} + 76A(A_r)^2C^3C_{rr} + 24A^3A_{tt}C^3C_{rr}
\end{aligned}
$$

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$$- 4 A^2 (A_t)^2 C^3 C_{r\,r} - 20 A^3 (C_r)^4 - 28 A^2 A_r C (C_r)^3 + 8 A^4 C C_{t\,t} (C_r)^2$$

$$+ 24 A^4 (C_t)^2 (C_r)^2 + 4 A^3 A_t C C_t (C_r)^2 + 12 A^2 A_{r\,r} C^2 (C_r)^2$$

$$- 31 A (A_r)^2 C^2 (C_r)^2 - 8 A^2 (A_t)^2 C^2 (C_r)^2 - 8 A^4 C^2 C_{t\,t\,r} C_r$$

$$+ 4 A^3 A_r C^2 C_{t\,t} C_r - 64 A^4 C C_t C_{t\,r} C_r + 12 A^3 A_t C^2 C_{t\,r} C_r$$

$$+ 4 A^3 A_r C (C_t)^2 C_r + 14 A^2 A_t A_r C^2 C_t C_r - 8 A^3 A_{t\,r} C^2 C_t C_r$$

$$- 8 A^2 A_{r\,r\,r} C^3 C_r + 52 A A_r A_{r\,r} C^3 C_r - 56 (A_r)^3 C^3 C_r - 52 A^2 A_{t\,t} A_r C^3 C_r$$

$$+ 38 A (A_t)^2 A_r C^3 C_r + 40 A^3 A_{t\,t\,r} C^3 C_r - 36 A^2 A_t A_{t\,r} C^3 C_r$$

$$- 8 A^5 C^2 C_t C_{t\,t\,t} + 8 A^4 A_t C^3 C_{t\,t\,t} - 16 A^4 C^3 C_{t\,t\,r\,r} + 8 A^3 A_r C^3 C_{t\,t\,r}$$

$$+ 4 A^5 C^2 (C_{t\,t})^2 + 8 A^5 C (C_t)^2 C_{t\,t} - 16 A^4 A_t C^2 C_t C_{t\,t} - 8 A^4 A_{t\,t} C^3 C_{t\,t}$$

$$+ 8 A^3 (A_t)^2 C^3 C_{t\,t} + 40 A^4 C^2 C_t C_{t\,r\,r} - 16 A^3 A_t C^3 C_{t\,r\,r}$$

$$+ 32 A^4 C^2 (C_{t\,r})^2 - 20 A^3 A_r C^2 C_t C_{t\,r} + 24 A^2 A_t A_r C^3 C_{t\,r}$$

$$- 16 A^3 A_{t\,r} C^3 C_{t\,r} - 4 A^5 (C_t)^4 + 4 A^4 A_t C (C_t)^3 + 12 A^4 A_{t\,t} C^2 (C_t)^2$$

$$- 3 A^3 (A_t)^2 C^2 (C_t)^2 + 8 A^2 A_t A_{r\,r} C^3 C_t - 20 A A_t (A_r)^2 C^3 C_t$$

$$+ 20 A^2 A_{t\,r} A_r C^3 C_t + 8 A^4 A_{t\,t\,t} C^3 C_t - 24 A^3 A_t A_{t\,t} C^3 C_t$$

$$- 8 A^3 A_{t\,r\,r} C^3 C_t + 10 A^2 (A_t)^3 C^3 C_t - 16 A^2 A_{t\,t} A_{r\,r} C^4$$

$$+ 16 A (A_t)^2 A_{r\,r} C^4 + 40 A A_{t\,t} (A_r)^2 C^4 - 56 (A_t)^2 (A_r)^2 C^4$$

$$- 40 A^2 A_{t\,t\,r} A_r C^4 + 72 A A_t A_{t\,r} A_r C^4 - 8 A^3 A_t A_{t\,t\,t} C^4$$

$$+ 16 A^3 A_{t\,t\,r\,r} C^4 + 4 A^3 (A_{t\,t})^2 C^4 + 12 A^2 (A_t)^2 A_{t\,t} C^4$$

$$- 16 A^2 A_t A_{t\,r\,r} C^4 - 16 A^2 (A_{t\,r})^2 C^4 - 7 A (A_t)^4 C^4 - 16 A^5 C^2)/(48 A^5 C^4)$$

5. ISOTROPIC, COMOVING COORDINATES

In this case the line element is given by

$$ds^2 = -dt^2 + A(r, t)(dr^2 + r^2 d\Omega) ,$$

and the relevant components of $W^{\mu\nu}$ are denoted by F^{rr}, F^{rt}, and F^{tt}. They are found to take the forms which follow below. In this particular case we note that when we set $A(r, t) = R^2(t)/(1 - kr^2)^2$, $F^{\mu\nu}$ is found to vanish identically, as is to be expected since a Friedmann-Robertson-Walker universe is conformal to Minkowski spacetime.

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$$F^{rr} =$$

$$(8A^2 A_r A_{r\,r\,r}\, r^3 - 4A^2 (A_{r\,r})^2 r^3 - 20A (A_r)^2 A_{r\,r}\, r^3$$

$$- 16A^3 A_{t\,t} A_{r\,r}\, r^3 + 24A^2 (A_t)^2 A_{r\,r}\, r^3 + 15 (A_r)^4 r^3 + 48A^2 A_{t\,t} (A_r)^2 r^3$$

$$- 120A (A_t)^2 (A_r)^2 r^3 - 48A^3 A_{t\,t\,r} A_r r^3 + 168A^2 A_t A_{t\,r} A_r r^3$$

$$+ 16A^4 A_{t\,t\,r\,r}\, r^3 - 24A^3 A_t A_{t\,r\,r}\, r^3 - 48A^3 (A_{t\,r})^2 r^3 + 16A^3 A_{r\,r\,r}\, r^2$$

$$- 48A^2 A_r A_{r\,r}\, r^2 + 20A (A_r)^3 r^2 + 16A^3 A_{t\,t} A_r r^2 - 24A^2 (A_t)^2 A_r r^2$$

$$- 16A^4 A_{t\,t\,r}\, r^2 + 24A^3 A_t A_{t\,r}\, r^2 + 32A^3 A_{r\,r}\, r - 68A^2 (A_r)^2 r - 32A^3 A_r)$$

$$/(48A^7 r^3)$$

$$F^{rt} =$$

$$(2A^2 A_t A_{r\,r\,r}\, r^2 - 15A A_t A_r A_{r\,r}\, r^2 + 8A^2 A_{t\,r} A_{r\,r}\, r^2$$

$$+ 15A_t (A_r)^3 r^2 - 15A A_{t\,r} (A_r)^2 r^2 + 7A^2 A_{t\,r\,r} A_r r^2 - 2A^3 A_{t\,r\,r\,r}\, r^2$$

$$+ 4A^2 A_t A_{r\,r}\, r - 15A A_t (A_r)^2 r + 15A^2 A_{t\,r} A_r r - 4A^3 A_{t\,r\,r}\, r$$

$$- 4A^2 A_t A_r + 4A^3 A_{t\,r})/(6A^6 r^2)$$

$$F^{tt} =$$

$$(16A^3 A_{r\,r\,r\,r}\, r^2 - 80A^2 A_r A_{r\,r\,r}\, r^2 - 60A^2 (A_{r\,r})^2 r^2$$

$$+ 260A (A_r)^2 A_{r\,r}\, r^2 - 135 (A_r)^4 r^2 + 64A^3 A_{r\,r\,r}\, r - 280A^2 A_r A_{r\,r}\, r$$

$$+ 220A (A_r)^3 r - 20A^2 (A_r)^2)/(48A^6 r^2)$$

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